

# Pragmatic factors in the processing of Mandarin focus intervention effects

Jess H.-K. Law<sup>1</sup>, Haoze Li<sup>2</sup>, and Yaqing Cao<sup>3</sup>

<sup>1</sup>*University of California, Santa Cruz*

<sup>2</sup>*Nanyang Technological University, Singapore*

<sup>3</sup>*Peking University*

## Abstract

An in-situ *wh*-question is generally considered degraded if the *wh*-phrase is preceded by a focus phrase, a phenomenon known as ‘focus intervention effects’. In syntax and semantics, focus intervention effects have been considered crucial evidence for various grammatical approaches to *wh*-in-situ questions. However, the degradedness of sentences involving focus intervention effects has long been disputed, especially in the face of pragmatic manipulations on *wh*-phrases and focus phrases. The present study reports three experiments probing the role of two pragmatic factors in focus intervention effects: the givenness of a focus phrase and the linguistic informativity of a *wh*-phrase, using both acceptability judgment and self-paced reading tasks. The experimental results show that linguistic informativity does not influence focus intervention effects, whereas focus givenness leads to a facilitation effect in both tasks. However, the facilitation does not completely eliminate focus intervention effects. Based on these findings, we reaffirm the relevance of grammatical approaches to focus intervention effects developed in the syntax and the semantics tradition while also acknowledging the partial role of pragmatics.

## Contents

|          |                                                  |          |
|----------|--------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                              | <b>2</b> |
| <b>2</b> | <b>Setting up the debate</b>                     | <b>5</b> |
| 2.1      | Syntactic-semantic approach . . . . .            | 5        |
| 2.2      | Pragmatic approach . . . . .                     | 7        |
| 2.2.1    | Focus givenness . . . . .                        | 8        |
| 2.2.2    | D-linking and linguistic informativity . . . . . | 9        |
| 2.3      | Motivating an experimental approach . . . . .    | 11       |

|          |                                                                              |           |
|----------|------------------------------------------------------------------------------|-----------|
| <b>3</b> | <b>Experiment 1: Are FIEs eliminated by focus givenness?</b>                 | <b>12</b> |
| 3.1      | Stimuli . . . . .                                                            | 13        |
| 3.2      | Procedure . . . . .                                                          | 14        |
| 3.3      | Participants . . . . .                                                       | 15        |
| 3.4      | Data treatment . . . . .                                                     | 16        |
| 3.5      | Results . . . . .                                                            | 16        |
| 3.5.1    | Acceptability ratings . . . . .                                              | 16        |
| 3.5.2    | Reading times . . . . .                                                      | 18        |
| 3.6      | Summary . . . . .                                                            | 20        |
| <b>4</b> | <b>Experiment 2 and 3: Does <i>wh</i>-informativity play a role in FIEs?</b> | <b>21</b> |
| 4.1      | Experiment 2: An acceptability judgment task . . . . .                       | 21        |
| 4.1.1    | Stimuli . . . . .                                                            | 21        |
| 4.1.2    | Procedure . . . . .                                                          | 24        |
| 4.1.3    | Participants . . . . .                                                       | 24        |
| 4.1.4    | Data treatment . . . . .                                                     | 25        |
| 4.1.5    | Results . . . . .                                                            | 25        |
| 4.2      | Experiment 3: A self-paced reading task . . . . .                            | 27        |
| 4.2.1    | Stimuli . . . . .                                                            | 27        |
| 4.2.2    | Procedure . . . . .                                                          | 28        |
| 4.2.3    | Participants . . . . .                                                       | 28        |
| 4.2.4    | Data treatment . . . . .                                                     | 28        |
| 4.2.5    | Results . . . . .                                                            | 29        |
| 4.3      | Interim summary . . . . .                                                    | 30        |
| <b>5</b> | <b>General discussion and theoretical outlook</b>                            | <b>30</b> |
| <b>6</b> | <b>Conclusion</b>                                                            | <b>32</b> |

## 1 Introduction

*Wh*-questions are attested in almost all languages. Cross-linguistically, two primary strategies are employed to form *wh*-questions: *wh*-movement and *wh*-in-situ. In *wh*-movement languages like English, *wh*-phrases are displaced from their base-generated positions to the clause-initial position, as illustrated in (1). In contrast, in *wh*-in-situ languages like Mandarin, *wh*-phrases remain in their original syntactic positions, as shown in (2). Explaining why natural languages differ in the syntactic realization of *wh*-questions and how these different strategies are related remains a central issue in contemporary linguistics (Chomsky 1981; Huang 1982; Cheng 1991; Li 1992; Tsai 1994; Richards 2001; Dong 2009; Cable 2010; Kotek 2014; a.o.).<sup>1</sup>

- (1) *Wh*-movement (English; also Malagasy, Mixtec, Paiwan, Russian, Tagalog, etc)
- What** did you order \_\_\_\_\_?
  - Where** did Sue go \_\_\_\_\_?

<sup>1</sup> Abbreviations used in the glosses of examples include: PFV = perfective marker; CLF = classifiers; EMP = emphatic marker; SFP = sentence final particle; MOD = modification marker.

- (2) *Wh*-in-situ (Mandarin; also Hindi, Hawaiian, Japanese, Pomo, Thai, Turkish etc)
- a. Nǐ mǎi-le **shěnmē dōngxī**?  
you buy-PFV what thing  
'What did you buy?'
  - b. Xiǎosū qù-le **nǎ'ēr**?  
Xiaosu go-PFV where  
'Where did Xiaosu go?'

Following Huang (1982), Watanabe (1991), Cheng (1991), Aoun and Li (1993), Shi (1994); Tsai (1994), and many others, it is argued that *wh*-movement and *wh*-in-situ involve **two types of interrogative dependencies** with the same interpretive structure: for *wh*-movement, a dependency is established between a moved *wh*-phrase and the gap in its base-generated position; while, for *wh*-in-situ, a dependency is formed between an interrogative operator in the sentence-initial position and an in-situ *wh*-phrase, which is often treated as a variable or alternative-evoking element. An important argument for positing interrogative dependencies in both *wh*-movement and *wh*-in-situ comes from their sensitivity to **locality effects**.

In *wh*-movement languages, it is well known that interrogative dependencies are sensitive to **syntactic islands** (Ross 1967; May 1977; Chomsky 1981, a.o.). For example, the dependencies involving *what* and *where* are ill-formed because *wh*-movement has taken place from a complex DP island in (3-a) and an adjunct island in (3-b).

- (3) a. \***What** did you praise [<sub>Island</sub> a friend that ordered \_\_\_\_]?  
b. \***Where** did Sue meet a new friend [<sub>Island</sub> after she worked \_\_\_\_]?

In *wh*-in-situ languages, it has been found that in-situ *wh*-nominals, as opposed to *wh*-adverbials, are largely subject not to syntactic islands (Huang 1982; Cheng 1991; Tsai 1994; Tian et al. 2022 cf. Lu et al. 2020), but rather to 'intervention effects': the acceptability of a *wh*-question is degraded once the *wh*-phrase is preceded by a certain expression (Beck 1996, 2006; Beck and Kim 2006; Yang 2008, 2012). Two major types of interveners have been identified in the literature—quantificational phrases and focused phrases—as illustrated in the examples below. Example (4) illustrates a quantifier intervention effect, where the intervener is a negative quantifier. In contrast, (5) and (6) exemplify focus intervention effects, with the interveners being focused phrases associated with focus particles such as *zhǐyǒu* only and *shì* (an emphatic marker). Focused phrases are capitalized in all relevant examples in this paper. When the *wh*-phrases are fronted over the interveners, the sentences become acceptable.<sup>2</sup>

- (4) a. ?\**Měiyǒu* rén chī-guò **shěnmē dōngxī**?  
not person eat-PFV what thing  
b. **Shěnmē dōngxī** *měiyǒu* rén chī-guò?  
what thing not person eat-PFV  
'What is the food *x* such that no one has eaten *x*?'
- (5) a. ?\**Zhǐyǒu* Nǐ mǎi-le **shěnmē dōngxī**?  
only you buy-PFV what thing

<sup>2</sup> As a reviewer points out, focus and *wh*-phrases cannot co-occur in many languages (Aboh 2007; Rizzi and Bocci 2017). However, FIEs do not arise from mutual incompatibility between focus and *wh*-phrases, since *wh*-phrases can precede focus phrases within a sentence.

- b. **Shěnme dōngxi** *zhǐyǒu* Nǐ mǎi-le?  
 what thing only you buy-PFV  
 ‘What is the thing *x* such that only you bought *x*.’
- (6) a. **?\*Shì** XIǎOZHĀNG fùzé-de **nǎ-ge** **xiàngmù**?  
 EMPH Xiaozhang take.charge.of-PFV which-CLF project  
 b. **Nǎ-ge** **xiàngmù** shì XIǎOZHĀNG fùzé-de?  
 which-CLF project EMPH Xiaozhang take.charge.of  
 ‘What is the project *x* such that it is Xiaozhang who takes charge of *x*?’

Throughout this paper, we use the notation ‘?’ to mark a shared intuitive judgment among native speakers: although the intended meaning of an intervention sentence can be recovered in context, the sentence itself is not a natural or acceptable way to express that meaning. This judgment differs from absolute ungrammaticality and aligns with the widely observed cross-linguistic pattern of FIE sentences, namely, that the degradation they trigger is subtler than that of clearly ungrammatical sentences (e.g., those involving incorrect word order).

Although there have been attempts to provide a unified account of quantifier and focus intervention effects, the types of quantifiers that trigger intervention vary cross-linguistically. For example, universal quantifiers are typical interveners in German but not in Mandarin. Li and Law (2016) argue that such variability reflects the heterogeneous nature of quantifier intervention effects and motivates language-specific analyses. By contrast, focus interveners exhibit much greater uniformity across languages. As a result, the study of FIEs offers a clearer window into a potentially universal mechanism underlying the formation of interrogative dependencies in *wh*-in-situ languages. For this reason, the present paper concentrates on focus intervention effects (henceforth FIE) and leaves quantifier intervention effects to future research.<sup>3</sup>

Locality effects in interrogative dependencies are landmark discoveries in the research on the syntax-semantics interface. A large number of syntactic and semantic theories are built to account for these locality conditions. However, the current syntactic framework—the Minimalist Program, has initiated a rethinking of locality conditions on structural dependencies as a so-called ‘third factor’ in language design (Chomsky 2005), which could be an extra-grammatical, domain-general cognitive/biological principle. So far, the extra-grammatical factors that have been explored include memory limitations (Kluender and Kutas 1993; Hofmeister 2007; Sprouse et al. 2013b), verb-frame frequencies (Liu et al. 2022), as well as information structural factors (Erteschik-Shir 1973; Kuno 1987; Goldberg 2013).

Experimental research has become crucial for assessing the extent to which extra-grammatical factors can account for locality effects. In studies of syntactic islands, Sprouse et al. (2012, 2016) argue that these effects are at least partially grammatical in nature, while Liu et al. (2022) advocates an explanation grounded primarily in extra-grammatical factors for most island phenomena, though acknowledges that a purely extra-grammatical account remains *speculative*.

In the case of FIEs, the picture remains far less clear, largely due to the scarcity of experimental studies. As with island effects, two major lines of analysis have been proposed: grammatical and pragmatic. Under grammatical approaches, FIEs are attributed to violations of syntactic or semantic constraints, thereby reflecting the grammatical derivation of interrogative dependencies.

<sup>3</sup> It is worth noting that, given the uniformity of FIEs, prominent unified approaches often analyze quantifier intervention effects as reducible to FIEs (Kim 2006; Beck 2006).

In contrast, pragmatic approaches view FIEs as arising from focus competition within information structure. These two perspectives yield distinct predictions on contexts. To date, the only experimental study we are aware of is Jin and Yan (2024), which finds that discourse linking does not affect FIEs, while leaving open the role of other potential pragmatic factors.

The goal of this article is to report experimental evidence that adjudicates between the grammatical view and a particular extra-grammatical view for FIEs, one that attributes them to **pragmatics**. Based on four experiments probing two pragmatic factors, namely, *wh*-informativity and focus givenness, we argue that **while pragmatics reduces the strength of degradedness associated with FIEs, FIEs do not reduce entirely to pragmatics**. Based on this finding, we conclude that FIEs, just like syntactic islands, also at least partially reside within a part of natural language grammar that is disjoint from pragmatics.

The remainder of this article is organized to support this conclusion. In section 2, we synthesize the literature on FIEs and group extant studies into two broad camps—the grammatical camp (section 2.1) and the pragmatic camp (section 2.2). We highlight the relevance of two factors that have been claimed to modulate FIEs, namely **focus givenness** and **linguistic informativity**. In section 2.3, we formulate two hypotheses based on these two factors. We then report three experiments aiming to test the focus givenness hypothesis (Section 3) and the linguistic informativity hypothesis (section 4.2). The experimental results show that these pragmatic factors are useful but not powerful enough to eliminate FIEs. We discuss how the implications of the experimental results in section 5, before concluding in section 6.

## 2 Setting up the debate

There is a general consensus among scholars working on FIEs that they follow from interactions between focus computation and question computation. Some scholars also agree that the interactions arise because focus and interrogative *wh*-phrases are *similar* in important respects (Beck 1996, 2006; Beck and Kim 2006; Tomioka 2007; Yang 2008; Eilam 2011). However, since both focus and questions have properties spanning across phonological, syntactic, semantic, and pragmatic dimensions, scholars disagree on which dimension (or dimensions) to understand the interactions leading to FIEs. An important divide in the disagreement lies in the relevance of pragmatics in FIEs:

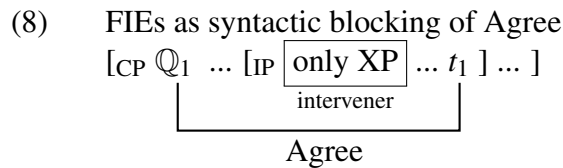
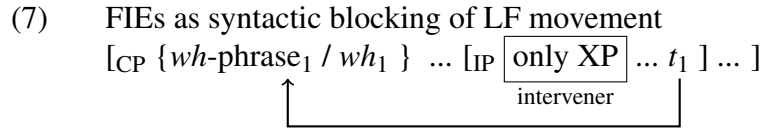
- the syntax-semantic view: FIEs result from violating syntax or semantic principles, which govern the well-formedness of *sentences*.
- the pragmatic view: FIEs involve violation of pragmatic principles, which govern not the well-formedness of sentences, but the well-formedness of *sentence uses* in context.

In this section, we will outline the main ideas of these two views and elaborate on their empirical implications to motivate the present experimental study. Theoretical details that do not bear on the empirical predictions tested are largely omitted.

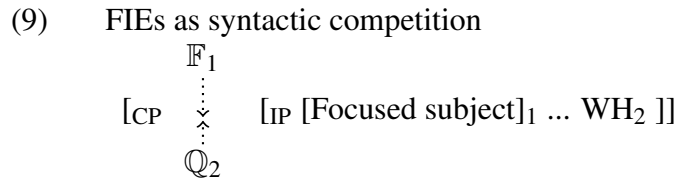
### 2.1 Syntactic-semantic approach

Under the syntax–semantics approach, the interaction between focus and *wh*-phrases yields structural ill-formedness, giving rise to FIEs independently of context. On the syntactic side, such

ill-formedness has been modeled in terms of feature-driven blocking or operator competition. In blocking accounts, the syntactic operation that links an in-situ *wh*-phrase to the scope position responsible for interrogative interpretation is obstructed. This operation may involve covert movement (Beck 1996; Beck and Kim 1997) or feature movement (Soh 2005) of the *wh*-phrase, as shown in (7). Besides movement, the operation may also be formalized as an Agree relation between an interrogative operator and an in-situ *wh*-phrase, as illustrated in (8) (Kim 2006; Li 2011; Li and Cheung 2015).



In the operator competition account, focus and question operators,  $\mathbb{F}$  and  $\mathbb{Q}$  respectively, vie for the same syntactic position, as depicted in (9) (Yang 2008, 2012).<sup>4</sup>

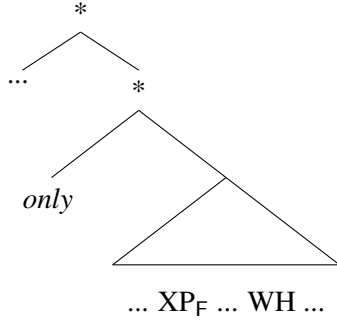


To account for the canceling effect of *wh*-fronting, this competition account proposes that the question operator binding the fronted *wh*-phrase occupies a position higher than the focus operator. Specifically, once a *wh*-phrase is fronted over a focus phrase and its associated focus operator, a question operator must be merged in a higher position to bind the *wh*-phrase. As a result, the question operator must also be higher than the focus operator. This follows from the cartographic view of the left periphery, which assumes multiple functional projections (Rizzi 1997). Hence, *wh*-fronting constructions do not involve competition between focus and question operators.

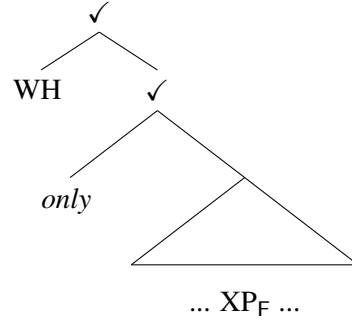
On the semantic side, FIEs are taken to involve failure at the level of semantic composition or violation of interpretive requirements. According to Beck (2006) and subsequent studies, FIEs reflect a failure in composing a focus-sensitive particle like *only* with a constituent containing both a focus phrase and a *wh*-phrase (Kotek 2014; Kotek and Erlewine 2016; Li and Law 2016; Erlewine 2025; a.o.), as illustrated in (10). Once the *wh*-expression is located outside the surface scope of the focus-sensitive particle, no intervention effects arise, as shown in (11). Note that the mother nodes represent the predicted well-formedness of the associated constituents.

<sup>4</sup> Strictly speaking, Yang's analysis suggests that FIEs arise from a competition effect rather than a genuine intervention effect, although they superficially resemble intervention patterns in surface structure.

(10)



(11)



Mayr (2014) proposes another semantic analysis, in which FIEs are caused by a semantic conflict between focus interveners and question interpretation. In particular, he notices that a focus intervener like *only* is non-additive as the expression  $\llbracket \text{only} \rrbracket(p \vee q)$  is not truth-conditionally equivalent to the expression  $\llbracket \text{only} \rrbracket(p) \vee \llbracket \text{only} \rrbracket(q)$  (e.g., *only SUE attended the dinner or the ball* vs. *only SUE attended the dinner or only SUE attended the ball*). However, as Mayr posits, a *wh*-question has a semantic requirement that the disjunction of its possible answers should return its existential presupposition where the *wh*-phrase is interpreted as an existential quantifier. Let's take (5), repeated below as (12), as an example. Its possible answers would be the ones included in the set shown in (13). Disjoining them gives rise to a different truth condition from the existential presupposition of this question due to the presence of *only*. In the scenario that Libai is the only person who bought A, but others bought B, the disjunction on the left-hand side of (13) is true, whereas the proposition on the right-hand side is not.

(12) ?\*Zhǐyǒu Nǐ mǎi-le shěnmē dōngxi?

only you buy-PFV what thing

'What is the thing  $x$  such that only you bought  $x$ .'

$$(13) \quad \vee \left\{ \begin{array}{l} \llbracket \text{only YOU bought A} \rrbracket, \\ \llbracket \text{only YOU bought B} \rrbracket, \\ \llbracket \text{only YOU bought C} \rrbracket \end{array} \right\} \neq \llbracket \text{only} \rrbracket \left( \exists x \in \left\{ \begin{array}{l} \llbracket A \rrbracket, \\ \llbracket B \rrbracket, \\ \llbracket C \rrbracket \end{array} \right\} : \llbracket \text{YOU bought } t_1 \rrbracket^{g^1 \rightarrow x} \right)$$

The in-equivalence in (13) arises from the scope of the *wh*-phrase within the focus particle *only*. As a result, once *wh*-phrases are fronted beyond focus particles, FIEs are expected to vanish.

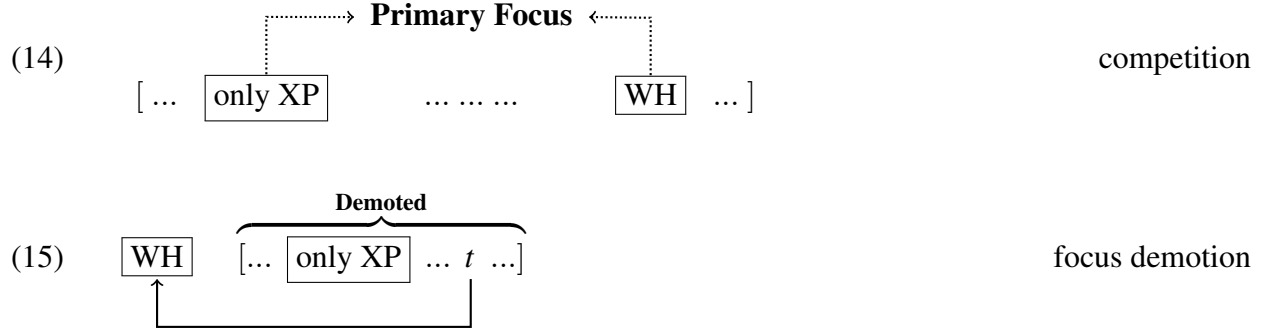
Despite their differences, the accounts in the syntax-semantics approach all take FIEs to reflect a defect at the level of sentence computation, which is independent of the use of a sentence. As a prediction, the context in which the relevant sentence is uttered plays little or no role in improving FIEs. This prediction is challenged by the pragmatic approach, to which we turn next.

## 2.2 Pragmatic approach

Under the pragmatic approach, however, FIEs occur not because the relevant *wh*-question is problematic at the sentence level, but because it is problematic in context. This view predicts that manipulations of context, or of aspects that directly relate to context, can remedy or even eliminate FIEs. Two such context-dependent factors have been highlighted in the literature: (i) given focus, where FIEs disappear once the focus phrase is contextually given; and (ii) D(iscourse)-linked *wh*-phrases, where FIEs are alleviated once the *wh*-domain is modified and becomes more informative.

### 2.2.1 Focus givenness

Tomioaka (2007) and Eilam (2011) argue that FIEs are triggered when a focus and a *wh*-phrase compete for the primary focus status in a *wh*-question exhibiting FIEs, as visualized in (14). To prevent the competition, the focus has to be demoted. Specifically, *wh*-fronting induces a form of structural demotion, whereby the material following the fronted *wh*-phrase undergoes prosodic reduction, as shown in (15).



This prosodic reduction affects information structure—put simply, the reduced portion can no longer bear the primary focus. Hence, in *wh*-fronting constructions, the fronted *wh*-phrase does not compete with any subsequent focus phrase for primary focus, and FIEs are therefore not expected.

Furthermore, Eilam (2011) goes as far as arguing that *contextually demoting* the focus status of a focused phrase, by rendering it given, for example, also ameliorates FIEs. According to this view, (16) can become acceptable if it is uttered following the context in (30), which entails that *Lili* is the only individual that bought a book.<sup>5</sup> The context turns the focus phrase into a piece of given information. As a consequence, it cannot be used as the primary focus, which must be novel in information structure, and then does not compete with the *wh*-phrase. This observation is also reported for French, in Glasbergen (2021).<sup>6</sup>

(16) Context: The class was assigned two book reports. Lili read one book that everyone else had read, but there was one book that only Lili had read.

S: (?)Zhǐyǒu LÌLÌ dú-le nǎ-bě'n shū?  
only Lili read-PFV which-CLF book

‘What is the book *x* such that only LILI read *x*?’ (Eilam 2011: 70-71)

It is worth noting that a given phrase a given phrase can still function as the associate of a focus particle (Partee 1999; Ishihara and Féry 2006; Beaver and Clark 2008; among others). The phenomenon known as second occurrence focus provides independent evidence for this, as illustrated by the following conversation.

(17) A: Zhè-cì kǎoshì, Lǐbái zhǐ kào-le LIÙSHÍ fēn.  
this-CLF exam Libai only test-PFV 60 score

<sup>5</sup> The notation ‘(?)’ follows Eilam (2011) and indicates that the sentence is basically acceptable, though not fully natural.

<sup>6</sup> The contextual demotion strategy operates independently of the structural demotion strategy. The context in (16) does not typically trigger *wh*-fronting, which is generally motivated in situations where *wh*-phrases are used to inquire about the unique individual in a D-linked domain (Wu 1999; Pan 2014; Cheung 2014).

‘Libai got only 60 in this exam.’

- B: Zhèngcháng. *Lián DUFŭ dōu zhǐ kào-le LIÙSHÍ fēn.*  
normal even Dufu all only test-PFV 60 score  
‘It is not surprising. Even Dufu got only 60.’

In this example, the focus phrase in A’s utterance is repeated in B’s response and continues to associate with the focus particle.

For convenience of reference, the role of contextually given focus in rescuing FIEs is formulated into a hypothesis, stated in (18).

(18) **Focus Givenness Hypothesis**

- a. Strong: If the focus phrase in a focus intervention sentence is contextually given, FIEs disappear.
- b. Weak: If the focus phrase in a focus intervention sentence is contextually given, FIEs improve but may not entirely disappear.

Anticipating the possibility that focus givenness may have partial effects on FIEs, this hypothesis is split into a strong version and a weak version. The strong version predicts that FIEs should completely disappear if the intervening focus is given, while the weak version predicts that focus givenness improves FIEs, but does not necessarily eliminate them.

### 2.2.2 D-linking and linguistic informativity

Besides focus givenness, the role of D(iscourse)-linking is another pragmatic factor considered to interact with FIEs (Pesetsky 2000; Beck 2006; Jin and Yan 2024). D-linking is a particular type of informativity that has been often discussed. It refers to the situation that the witness set of a quantifier, such as an interrogative quantifier, is drawn from a set familiar to the conversational participants in a context (Pesetsky 1987). For example, the *wh*-phrase *what book* in (19) is not D-linked as there is no presupposition that the book John bought is drawn from a familiar set. However, the *wh*-phrase *which book* in (20) is D-linked as the book John bought has to be drawn from a set both the speaker and the addressee know about.

(19) **What book** did John buy? Non-D-linked

(20) **Which book** did John buy? D-linked

Pesetsky (2000) suggests that at least in *wh*-movement languages, D-linked *wh*-phrases, like *which*-NPs in English, undergo feature movement, which can be interrupted by a focus phrase. By contrast, non-D-linked *wh*-phrases undergo LF phrasal movement, even in English, which is not affected by a focus phrase.

However, Beck (2006) cautions that the data are not without controversy. Jin and Yan (2024) further shows, with experimental evidence from Mandarin, that there is no contrast between a D-linked *wh*-phrase and a non-D-linked one in terms of FIEs. In other words, (21) and (22) do not contrast in Mandarin, in contrast to Pesetsky’s claim about English.

- (21) ?\*Zhǐyǒu XǎOSŭ mǎi-le yī-běn shěnmē shū ne?  
only Xiaosu buy-PFV one-CLF what book Q

- ‘What book did only Xiaosu buy?’ Non-D-linked
- (22) ?\*Zhǐyǒu XIǎOSŪ mǎi-le **nǎ-běn** **shū** ne?  
 only Xiaosu buy-PFV which-CLF book Q  
 ‘Which book did only Xiaosu buy?’ D-linked

Although Jin and Yan’s findings suggest that D-linking does not modulate FIEs, it does not rule out the possibility that other ways of enhancing D-linking effects may modulate FIEs. In fact, there is extensive research in both the theoretical literature and the experimental literature suggesting that the presence of a *modifier* helps overcome locality conditions. For example, it has been argued in the theoretical literature that the presence of a D-linked determiner plus a modifier weakens strong crossover effects, as shown by the contrast in (23) (Wasow 1972; Falco 2007). In this paper, we refer to the informativity introduced by linguistic modifiers as ‘linguistic informativity’.

- (23) Strong Crossover (Wasow 1972: 120)
- a. \***Who**<sub>1</sub> did the woman he<sub>1</sub> loved betray t<sub>1</sub>?  
 b. ?**Which famous writer**<sub>1</sub> did the woman he<sub>1</sub> loved betray t<sub>1</sub>?

In addition, it has been suggested, in the psycholinguistic literature, that linguistic informativity may induce memory facilitation effect in the processing of long-distance dependencies (Hofmeister 2007). Concretely, Hofmeister (2007) finds that the verb *discuss* was read faster in (25) than in (24), which indicates an effect of D-linking. In addition, there was a non-significant trend for faster reading times in the verb region in (26), suggesting the possibility of an additional effect of informativity via modification (Hofmeister 2007:1093–94).

- (24) **What** did the reporter that Scooter avoided **discuss** during an evening news segment?  
 (25) **Which poll** did the reporter that Scooter avoided **discuss** during an evening news segment?  
 (26) **Which political poll** did the reporter that Scooter avoided **discuss** during an evening news segment?

The generality of linguistic informativity in conditioning long-distance dependencies is supported by two additional observations. First, Hofmeister also observes a trend for linguistic informativity advantage in cleft sentences, suggesting that increased linguistic informativity may be a general facilitating factor conditioning the processing of ex-situ long-distance dependencies. Second, Shlonsky et al. (2020) observes that long-distance *wh*-in-situ dependencies in English and French multiple-*wh* questions also interact with linguistic informativity. Specifically, moving a D-linked *wh*-phrase (underlined) across a non-D-linked *wh*-phrase, in the so-called Inclusion configuration, shown in (27-b), is slightly degraded. However, it is less degraded than moving a non-D-linked *wh*-phrase over another non-D-linked *wh*-phrase, as in the Identity configuration, shown in (27-a), or moving a non-D-linked *wh*-phrase over a D-linked *wh*-phrase, as in the Inclusion configuration, shown in (27-c). This three-way contrast is argued to follow from Relativized Minimality (Rizzi 1990).

- (27) *Wh*-island violation
- a. ??**What** do you wonder **who** bought \_\_\_? Identity

- |                                                                  |                   |
|------------------------------------------------------------------|-------------------|
| b. ? <b>Which book</b> do you wonder <b>who</b> bought ____?     | Inclusion         |
| c. * <b>What</b> do you wonder <b>which student</b> bought ____? | Inverse inclusion |

Interestingly, Hofmeister (2007) and Hofmeister et al. (2013) observe that this pattern is *reversed* in Superiority violation, which involves the overt movement of a *wh*-object, a *wh*-phrase in the object position, over a *wh*-subject, a *wh*-phrase in the subject position.

(28) Superiority violation

- |                                                    |                   |
|----------------------------------------------------|-------------------|
| a. *Mary wondered <b>what who</b> read ____.       | Identity          |
| b. ?Mary wondered <b>what which boy</b> read ____. | Inverse Inclusion |
| c. *Mary wondered <b>which book who</b> read ____. | Inclusion         |

The reversal has led Shlonsky et al. (2020) to conclude that *wh*-in-situ in Superiority violation also involves *wh*-movement, albeit a *covert* one. According to them, the *wh*-subjects in (28) all have to undergo covert movement across a moved *wh*-object, forming intersecting chains. In the case of Inverse Inclusion in (28-b), the movement is more acceptable because it involves the movement of a more informative *wh*-phrase over a less informative one, which respects Relativized Minimality. If their conclusion is on the right track, then linguistic informativity not only modulates *wh*-movement, but also *wh*-in-situ, at least in English and French.

Mandarin *wh*-in-situ questions have also been argued to involve covert dependencies and yield behavioral patterns similar to those associated with the processing of overt dependencies (Xiang et al. 2013, Xiang et al. 2015). It is therefore reasonable to ask whether linguistic informativity also influences the acceptability and processing of *wh*-in-situ questions, and whether such effects, if present, interact with FIEs. We formulate two versions of the Linguistic Informativity Hypothesis below:

(29) **Linguistic Informativity Hypothesis**

- a. General version: Increased linguistic informativity of a *wh*-phrase facilitates the acceptability and processing of *wh*-in-situ questions generally, with or without FIEs.
- b. Specific version: Increased linguistic informativity of a *wh*-phrase has a specific ameliorating effect on FIEs.

### 2.3 Motivating an experimental approach

Acceptability judgments of linguistic data collected from a few native speakers, often also known as introspective data, have been widely used in generative linguistics. Sprouse et al. (2013a) show that introspective data published in *Linguistic Inquiry*, a leading journal in generative linguistics, are highly replicable. Chen et al. (2020) show that the same is true for introspective data in *The Syntax of Chinese* (Huang et al. 2009), a widely adopted textbook in Chinese linguistics. However, despite the effectiveness of using introspective data to answer many important theoretical questions, it is desirable to use an experimental method to investigate the role of pragmatics in FIEs, for three reasons: (i) FIEs exhibit gradient judgments with inter-speaker variability, (ii) the need to quantify the role of pragmatics to understand the nature of FIEs, and (iii) the need to better understand how FIEs are processed. These reasons are elaborated below.

First, FIEs were known to be a gradient phenomenon with considerable inter-speaker variability across languages, ever since their discovery by Hoji (1985). The variability has been noted for

Japanese (Hoji 1985; Tomioka 2007; Tomioka et al. 2020), Korean (Beck and Kim 2006), and Mandarin (Yang 2008; Li 2011; Li and Law 2016). Given the gradient status and the inter-speaker variability, it is desirable to collect quantitative data from a larger sample of the population to investigate how stable FIEs are as a phenomenon. This prompted Jin and Yan (2024) to conduct a series of acceptability judgment experiments to quantify the degradedness of FIEs. They showed that FIEs are indeed degraded, despite inter-speaker variability. However, their experiments did not address the role of focus givenness and *wh*-informativity in FIEs, which is the primary research question of the present study.

Second, the debate between the syntax-semantics approach and the pragmatics approach stems from a disagreement on the acceptability of focus intervention sentences in the presence of context. Specifically, while Eilam (2011) and Glasbergen (2021) report that contextually given focus eliminates FIEs in Mandarin and French respectively, this kind of amelioration is argued to be insignificant in Kotek (2014). Since this is an empirical disagreement on the effect of context, it is a question that can be tested using systematically collected acceptability judgment data from a larger pool of speakers.

Third, both the syntactic-semantic approach and the pragmatic approach heavily implicates the role of sentence processing. For example, the syntactic approach predicts that factors relevant for dependencies should also be relevant for FIEs while the semantic approach and the pragmatic approach do not make these predictions. To make more fine-grained adjudications among these hypotheses, it is beneficial to learn how FIEs are processed. Reading times have been widely used to make inferences about cognitive processes. For example, reading time data are regularly used in the study of syntactic islands, leading to important discoveries about encoding and retrieval cost involved in these dependencies (see Liu et al. 2022 for a summary of important studies on this topic). We believe that data bearing on real-time processing offers an important window, though not an exclusive one, into the complex cognitive processes of interpreting sentences with FIEs relative to different types of contexts.

We report three experimental studies with acceptability judgment tasks and self-paced reading tasks, to test the Focus Givenness Hypothesis and the Linguistic Informativity Hypothesis, laid out earlier in this section. These studies are reported in section 3 (Experiments 1, on focus givenness) and section 4.2 (Experiments 2 and 3, on linguistic informativity).

### 3 Experiment 1: Are FIEs eliminated by focus givenness?

Our investigation starts from testing the Focus Givenness Hypothesis, which predicts that FIEs disappear when the focus that triggers FIEs is contextually given. The relevant example is repeated in (30), which Eilam takes to be unacceptable without the accompanying context.

- (30) Context: The class was assigned two book reports. Lili read one book that everyone else had read, but there was one book that only Lili had read.
- S: (?)Zhǐyǒu LÌLI dú-le nǎ-běn shū?  
only Lili read-PFV which-CLF book  
‘What is the book *x* such that only LILI read *x*?’ (Eilam 2011: 70-71)

To test this hypothesis, we manipulated the givenness status of the focus in an FIE sentence with the help of context. In particular, a focus phrase is *given* when the provided context entails its focus



**Figure 1:** A sample context used in Experiment 1

effect. When there is no context, the focus phrase is considered *novel*.<sup>7</sup>

### 3.1 Stimuli

This study used a  $3 \times 2$  design with two independent variables: *structure* (within-subjects) and *context* (between-subjects). A between-subject design for *context* was chosen because it allowed us to keep the trials consistent for each participant—a participant either saw all trials with context (in the PRESENT condition) or they saw all trials without context (in the ABSENT condition). A sample context used is shown in Figure 1. In this context, three people were introduced along with their names and pictures. The context entailed that only one person did something special. In this case, the special person was Xiaoqi. In a sample stimulus sentence, the proper name *Xiǎoqí* is marked as focus by *zhǐyǒu* ‘only’. Since both the exclusive interpretation and the alternatives to Xiaoqi had been presented in the context, the focus was considered given, as argued in (Eilam 2011; Hoeks et al. 2023). When there was no context, the focus was considered novel.

The second factor, *structure*, had three levels: INTERVENTION, PLAIN, and FRONTED. As in Experiments 1 and 2, INTERVENTION, exemplified in (31), was the target condition, with questions exhibiting FIEs. PLAIN was a control condition, with questions in the default word order not exhibiting FIEs, as shown in (32).<sup>8</sup>

- (31) *Zhǐyǒu* XIǎOQÍ mǎi-le **nǎ-běn** **zázhì** ne?  
 only Xiaoqi buy-PFV which-CLF magazine SFP  
 ‘Which magazine did Xiaoqi buy?’ INTERVENTION

<sup>7</sup> By probing into different types of contextual effects on FIE sentences, we may even be able to answer why and how context and FIEs interact at a more granular level, which can be helpful for shedding light on the nature of FIEs. We reserve this for future research.

<sup>8</sup> It is noted that the experimental items in this experiment include the interrogative sentence-final particle *ne*. To our knowledge, no existing research has suggested that this particle affects the occurrence of FIEs. Its inclusion here serves to enlarge the spillover region for comparing reaction times. Without the particle, the post-*wh* region would consist of only a single word—the nominal complement of the *wh*-determiner—which may be too short to capture potential differences in reaction times. A reviewer further pointed out that the particle does not appear in the stimuli used in Experiments 2 and 3. This is because those experiments were designed to test whether *wh*-modification influences FIEs, and their experimental materials were more complex, already containing sufficiently large spillover regions. Moreover, the difference in the inclusiveness of the interrogative particle does not affect the overall conclusions, as the results of Experiment 1 were not directly compared with those of Experiments 2 and 3.

| Condition   | structure    | context |
|-------------|--------------|---------|
| Condition 1 | PLAIN        | ABSENT  |
| Condition 2 | PLAIN        | PRESENT |
| Condition 3 | INTERVENTION | ABSENT  |
| Condition 4 | INTERVENTION | PRESENT |
| Condition 5 | FRONTED      | ABSENT  |
| Condition 6 | FRONTED      | PRESENT |

**Table 1:** Conditions in Experiment 1

- (32) Xiǎoqí mǎi-le **nǎ-běn zázhi** ne?  
 Xiaoqi buy-PFV which-CLF magazine SFP  
 ‘Which magazine did Xiaoqi buy?’ PLAIN

A potential concern is that the addition of focus may add an extra layer of complexity to sentences exhibiting FIEs, independently of FIEs. To better isolate the effect of FIEs independently of focus, an additional control condition, FRONTED, as exemplified in (33), was added. This configuration is known to encode focus but lack FIEs, because a *wh*-expression precedes the focus and its associate.

- (33) **Nǎ-běn zázhi**, zhǐyǒu XIǎOQÍ mǎi-le ne?  
 which-CLF magazine only Xiaoqi buy-PFV SFP  
 ‘Which magazine was it that only Xiaoqi bought it?’ FRONTED

Crossing the factors `context` and `structure` produced six conditions across two experiments, as shown in Table 1. Together, these conditions allow us to examine how different structures behave with and without contextual support. Specifically, if FIEs can be mitigated by focus givenness, the presence of a focus context should make a difference; otherwise, it should not. We constructed seven target items for each level of `structure`, yielding a total of 21 target stimuli. These were combined with 22 fillers and distributed evenly across three experimental lists, which were assigned to participants as a between-subject variable.

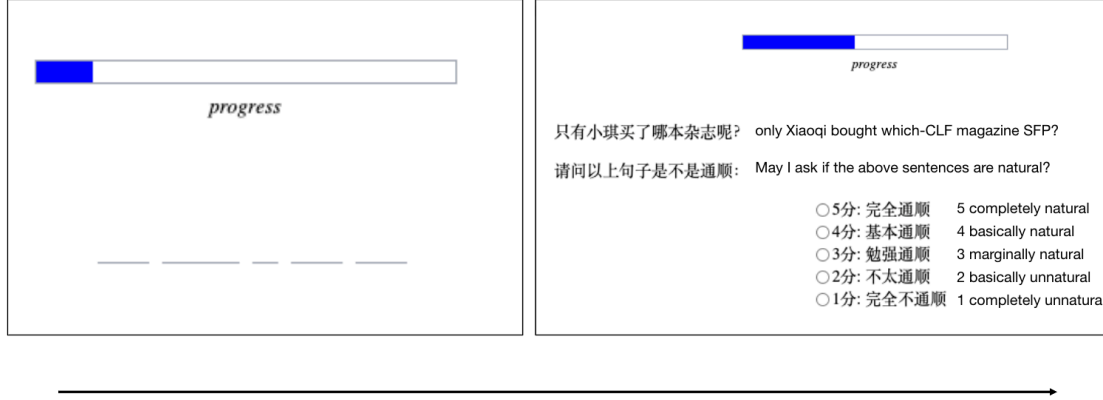
### 3.2 Procedure

The acceptability judgment task was designed and hosted on PCIBex. The entire experiment was presented in Simplified Chinese. In the instruction page, a participant first read a short text describing the experimental procedure, before being asked to provide their language background, age, and informed consent. This was followed by a practice phase, with three practice trials familiarizing a participant the procedure of an acceptability judgment task. The experiment stage began after the practice session. Each trial in the experimental stage proceeds in one of two ways, depending on the availability of context.

In the `context`-ABSENT condition, an experimental trial consisted of a self-paced reading task and an acceptability judgment task, with the former preceding the latter. In the self-paced reading task, a participant was instructed to press the space key to unveil the stimulus fragment by fragment, following their natural reading pace. An example of a sentence divided into six fragments is provided in Table 2. Response times between key presses were recorded. After the

|          |        |        |         |        |          |     |
|----------|--------|--------|---------|--------|----------|-----|
| Sequence | 1      | 2      | 3       | 4      | 5        | 6   |
| Fragment | zhiyou | Xiaoqi | mai-le  | shenme | zazhi    | ne? |
| Gloss    | only   | Xiaoqi | buy-PFV | what   | magazine | SFP |

**Table 2:** Fragments in the self-paced reading task in Experiment 1



**Figure 2:** Experimental procedure for the `context-ABSENT` condition

participant had completed the self-paced reading task, they then entered the acceptability judgment task. In this task, the same stimulus was displayed and the participant was directed to rate the acceptability of the sentence using a 5-point Likert scale. A one-minute break was given after half of the trials were completed. The procedure is illustrated in Figure 2, with English translations added for clarity in this paper.

A trial in the `context-PRESENT` condition was largely identical to that in the `ABSENT` condition, with the exception of two aspects. First, at the beginning of each trial, a context was shown and the participant was instructed to read it. In each context, a group of three individuals was described to be making decisions. This provides the focus alternative set, whose familiarity is known to influence the processing of focus constructions (Hoeks et al. 2023). The context showed that a particular individual, identified by a name, made a different decision from the rest of the group. This piece of information entails the focus contribution, rendering the focus given, which is an important manipulation for testing the Focus Givenness Hypothesis. Each participant got to decide how long they would like to engage with the context. After pressing a key to advance to the next step, a participant had to answer a comprehension question about the context, to verify that they had read the context. Feedback was given after each comprehension question, to emphasize the importance of reading the context. After the feedback was displayed, the rest of the procedure was identical to trials in the `context-ABSENT` condition. The procedure is shown in Figure 3. The sample test item is the same as in Figure 2, and the additional information not shown in Figure 2 is provided here with English translations.

### 3.3 Participants

A total of 311 participants were recruited from mainland China through personal networks. Among them, 148 took part in the `context-ABSENT` condition and 163 in the `context-PRESENT` con-



**Figure 3:** Experimental procedure for the `context-PRESENT` condition

dition. All participants were college students who had grown up in Mainland China and identified as native speakers of Mandarin. None of them was from a linguistics major and was reported to have any general cognitive impairments. Each participant received 20 CNY as compensation for their participation. Gender information was not collected.<sup>9</sup>

### 3.4 Data treatment

12 participants with an accuracy rate of 85% or lower for answering comprehension questions were removed. This affected about 4% of the data points in `context-PRESENT` condition. Any reading time that was slower than 100ms or longer than 1000ms was excluded.<sup>10</sup>

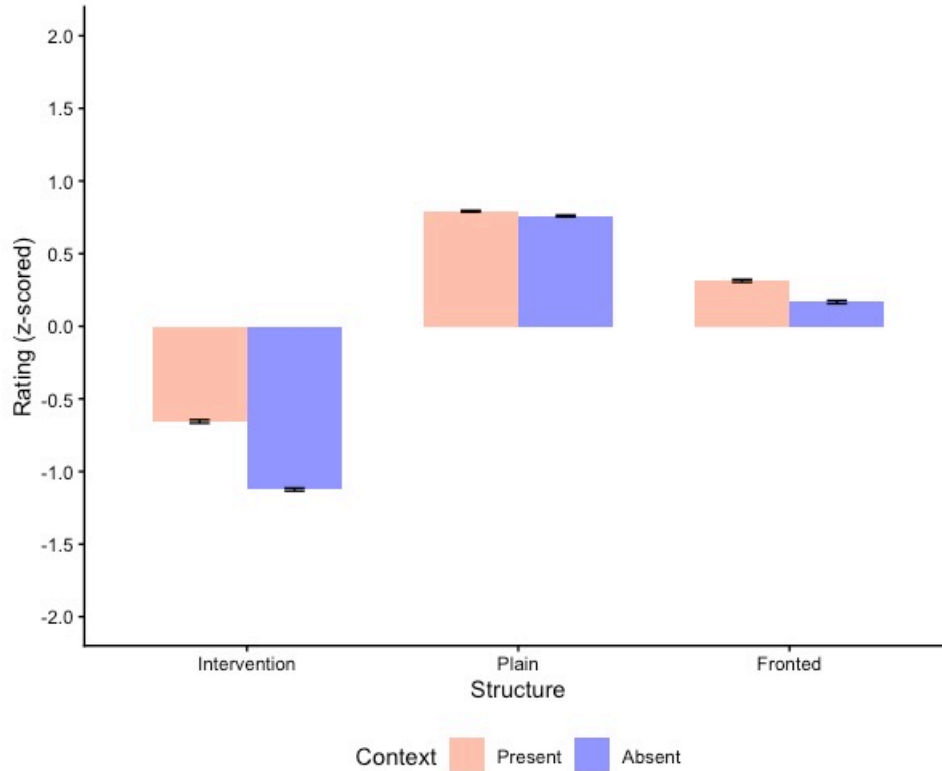
## 3.5 Results

### 3.5.1 Acceptability ratings

The 5-point ratings were first transformed to  $z$ -scores in order to control for individual bias in using the scale. The data, as summarized in Figure 4, were then analyzed using a linear mixed-effects model with `context` (`PRESENT` vs. `ABSENT`) and `structure` (`PLAIN`, `INTERVENTION`, `FRONTED`) as fixed effects, including their interaction. The results are shown in Table 3. `Context` was coded using sum contrasts (`PRESENT` = +1, `ABSENT` = -1). `Structure` was coded using two centered contrasts: the first compared `PLAIN` vs. `FRONTED`, and the second compared

<sup>9</sup> The recruitment process proceeded as follows. The authors first contacted colleagues and friends working at universities in Mainland China, who then disseminated the recruitment information and experiment link through their class WeChat groups. Interested students voluntarily signed up to participate. Upon completing the tasks, each participant received a reference number, which they submitted to claim their compensation. This procedure allowed us to monitor participants' backgrounds and ensure that all were native speakers of Mandarin and do not have cognitive impairments. Although the testing environments could not be controlled, comprehension questions were used in all experiments to ensure that participants attended to the the experiments.

<sup>10</sup> A few different cutoff ranges were experimented with but they did not yield substantial differences. This particular range was chosen because of simplicity.



**Figure 4:** Effects of structure and context on acceptability ratings (Experiment 1)

INTERVENTION vs. FRONTED, with the constraint that all contrasts summed to zero. This coding renders the model intercept interpretable as the grand mean across all experimental cells.

Figure 4 shows that the three levels of `structure` have distinct acceptability ratings. In PLAIN condition, ratings were the highest overall and only slightly affected by context (ABSENT:  $M = 0.77$ ,  $SE = 0.04$ ; PRESENT:  $M = 0.78$ ,  $SE = 0.04$ ). There was a significant main effect of context, such that ratings were higher in the PRESENT than in the ABSENT context ( $\beta = .10$ ,  $SE = .02$ ,  $t = 5.02$ ,  $p < .001$ ; corresponding to a PRESENT-ABSENT difference of .20). Significant deviations from the grand mean were observed for `structure`: ratings were higher for PLAIN ( $\beta = .73$ ,  $SE < .01$ ,  $t = 150.53$ ,  $p < .001$ ) and lower for INTERVENTION ( $\beta = -.93$ ,  $SE < .01$ ,  $t = -199.99$ ,

| Rating               | $\beta$ | $SE$ | $t$     | $p$ |
|----------------------|---------|------|---------|-----|
| Intercept            | .04     | .03  | 1.25    |     |
| Present              | .10     | .02  | 5.02    | *** |
| Plain                | .73     | .00  | 150.53  | *** |
| Intervention         | -.93    | .00  | -199.99 | *** |
| Present:Plain        | -.09    | .00  | -19.05  | *** |
| Present:Intervention | .13     | .00  | 27.62   | *** |

**Table 3:** Linear mixed-effects model of z-scored acceptability ratings with sum coded `structure` and `context`

| Rating               | $\beta$ | $SE$ | $z$     | $p$ |
|----------------------|---------|------|---------|-----|
| Present              | .37     | .08  | 4.42    | *** |
| Plain                | 2.75    | .02  | 105.18  | *** |
| Intervention         | -2.91   | .02  | -128.96 | *** |
| Present:Plain        | -.13    | .02  | -5.71   | *** |
| Present:Intervention | .26     | .01  | 14.78   | *** |

**Table 4:** Cumulative link mixed effects model of acceptability ratings with sum-coded `structure` and `context`

$p < .001$ ). The implied deviation for `FRONTED` was positive ( $\beta = .20$ ), reflecting its position above the grand mean.

The interaction between `context` and `structure` was also significant. When the contexts were present, the advantage of `PLAIN` over the grand mean was reduced when the contexts were present ( $\beta = -.09$ ,  $SE < .01$ ,  $t = 19.05$ ,  $p < .001$ ), whereas the `INTERVENTION` penalty was attenuated ( $\beta = .13$ ,  $SE < .01$ ,  $t = 27.62$ ,  $p < .001$ ). Correspondingly, model-predicted means showed higher ratings in the `context-PRESENT` condition than the `context-ABSENT` condition for all three levels of `structure`, with the largest contextual modulation observed for `INTERVENTION`.

To rule out concerns arising from the use of  $z$ -scored ratings with a linear mixed-effects model, we also fed the raw, non- $z$ -scored ratings to a cumulative-link (ordinal) mixed effects model to validate the conclusion. The results, shown in Table 4, are consistent with the results in Table 3.

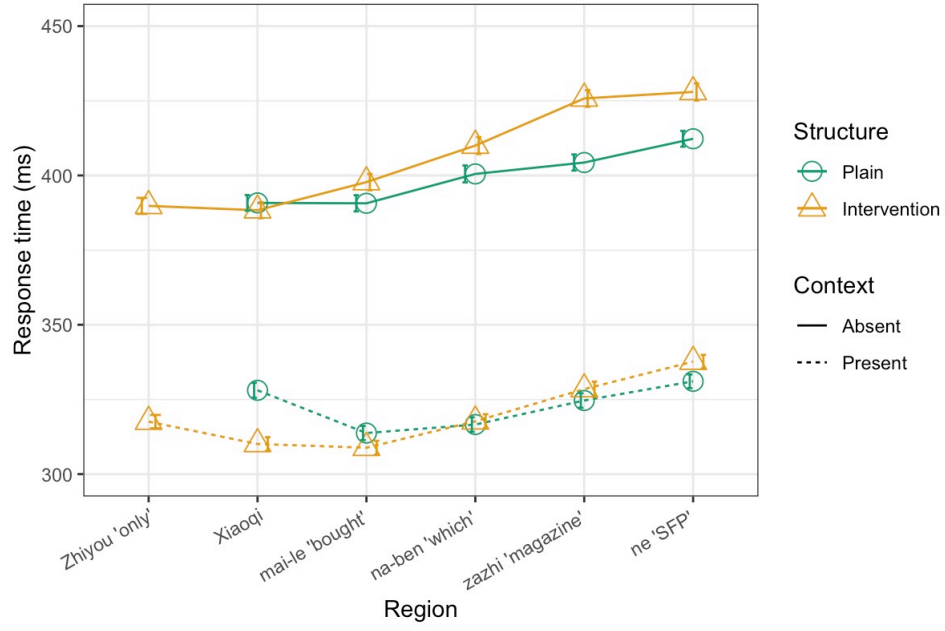
In sum, FIEs were robustly attested in an acceptability judgment task. Although this conclusion is by no means novel, as it has been informally reported in Kotek (2014) and more recently confirmed in Jin and Yan (2024), our study can be seen as providing corroborating evidence for the robustness of acceptability judgment tasks for detecting FIEs. In addition, although context plays some role in ameliorating the degradedness of FIEs, it does not suffice to fully neutralize FIEs. This is an important finding as it supports the weak version of the Focus Givenness Hypothesis but refutes its strong version.

### 3.5.2 Reading times

The data collected from the self-paced reading tasks in this experiment are visualized in the line plot in Figure 5. The reading times are log-transformed and analyzed using linear mixed-effects models with sum-coded (deviation-coded) contrasts for `Context` (`PRESENT` = +1, `ABSENT` = 1) and `Condition` (`PLAIN` = +1, `INTERVENTION` = 1), including their interaction. Under this coding, the intercept reflects the grand mean across the four cells, and each main-effect coefficient corresponds to half of the associated two-level difference.

Three linear mixed-effects models were fit to examine the effect of `structure`, `context`, and their interaction in the `WH`, `NP`, and `SFP` regions. The model results are reported in Table 5. Note that since the word order in the `FRONTED` condition was different from the word order in the other two conditions, this condition was not included in the by-region analysis.

Across regions, there was a robust main effect of `context`, with faster reading in the `PRESENT` than `ABSENT` context (`WH`:  $\beta = -.05$ ,  $SE = .01$ ,  $t = -5.62$ ,  $p < .001$ ; `NP`:  $\beta = -.05$ ,  $SE = .01$ ,  $t = -5.45$ ,  $p < .001$ ; `SFP`:  $\beta = -.05$ ,  $SE = .01$ ,  $t = -6.82$ ,  $p < .001$ ), corresponding to an estimated



**Figure 5:** Effects of structure and context on reading times

| WH                   | $\beta$ | SE  | t      | p   |
|----------------------|---------|-----|--------|-----|
| Intercept            | 2.51    | .01 | 272.51 | *** |
| Intervention         | .01     | .00 | 1.78   | .   |
| Present              | -.05    | .01 | -5.62  | *** |
| Intervention:Present | -.00    | .00 | -1.23  |     |
| NP                   | $\beta$ | SE  | t      | p   |
| Intercept            | 2.52    | .01 | 275.58 | *** |
| Intervention         | .01     | .00 | -3.44  | *** |
| Present              | -.05    | .01 | -5.45  | *** |
| Intervention:Present | -.01    | .00 | -2.67  | **  |
| SFP                  | $\beta$ | SE  | t      | p   |
| Intercept            | 2.54    | .01 | 333.59 | *** |
| Intervention         | .01     | .00 | 2.71   | **  |
| Present              | -.05    | .01 | -6.82  | *** |
| Intervention:Present | -.00    | .00 | -.45   |     |

**Table 5:** Linear mixed-effects model of log-transformed reading times with sum-coded structure and context

PRESENTABSENT difference of approximately .10 log units (i.e.,  $2 \times \beta$ ). This main effect is clearly visible in Figure 5.

Evidence for a `structure` effect was small: it was marginal at WH ( $\beta = .01$ ,  $SE < .01$ ,  $t = 1.78$ ,  $p = .075$ ), but significant at NP ( $\beta = .01$ ,  $SE < .01$ ,  $t = -3.44$ ,  $p < .001$ ) and SFP ( $\beta = .01$ ,  $SE < .01$ ,  $t = 2.71$ ,  $p < .01$ ). Finally, the `structure`  $\times$  `context` interaction was reliable only in the NP region ( $\beta = -.01$ ,  $SE < .014$ ,  $t = -2.67$ ,  $p < .01$ ), indicating that the magnitude of contextual facilitation (PRESENT vs. ABSENT) differed by condition at NP; no interaction was observed at WH ( $t = -1.23$ ,  $p > .1$ ) or SFP ( $t = 0.45$ ,  $p > .6$ ).

A facilitative role of context in focus processing has also been reported by Hoeks et al. (2023), who observe faster processing for focused sentences when both the focus and its alternatives are salient in the preceding discourse. This aligns with the classical view, originating with Roberts (2012)[1996], that prosodic focus presupposes an active question under discussion. Contexts that introduce both a focus and its alternatives thus both support this presupposition and anticipate an upcoming discourse move involving focus.

One might wonder whether such general context facilitation alone explains the absence of FIEs in the self-paced reading task when context was present. If so, the Focus Givenness Hypothesis would reduce to a specific instance of context facilitation, namely one that manifests as FIEs. This remains an open possibility. However, the facilitation reported by Hoeks et al. (2023) is relatively local, emerging only in the first few segments following the focus. It is therefore unclear whether this mechanism can account for the neutralization observed after the verb region in our data.

Furthermore, FIEs may re-emerge later, in the SFP region, as the SFP model shows a main effect of `structure` without a `structure`  $\times$  `context` interaction. This potential effect after the post-critical region is expected, as processing difficulty can spill over when the parser carries unresolved material forward (Just et al. 1982; Mitchell 1984). Hence, the measure in any given region may reflect not only the processing demands of that region itself but also any backlog accumulated in the preceding context.

Besides these main findings, we would like to make a remarks on interaction effects prior to the WH region, There was a significant interaction effect in the subject region (labeled with the example *Xiaoqi* in Figure 5;  $\beta = -.01$ ,  $SE < .01$ ,  $t = -2.27$ ,  $p < .05$ ) and a marginally significant interaction effect in the verb region (labeled with the example *mai-le* ‘bought’;  $\beta = -.01$ ,  $SE < .01$ ,  $t = -1.74$ ,  $p < .1$ ). This interaction effect suggests the relevance of context facilitation even before the critical WH region. In other words, when context was provided, the PRESENT condition was processed faster than the ABSENT condition very early on. The fact that there was no significant contrast between the two conditions in the subject region when there was no context further corroborates this point.

### 3.6 Summary

The context facilitation effect observed in both the acceptability rating and self-paced reading tasks suggests that focus givenness alleviates FIEs. However, because this effect only partially mitigates rather than completely eliminates FIEs, the results support only the weak version of the hypothesis. The strong version, as originally proposed by Eilam (2011), is not supported.

| Condition   | structure    | <i>wh</i> -modification |
|-------------|--------------|-------------------------|
| Condition 1 | PLAIN        | ABSENT                  |
| Condition 2 | PLAIN        | PRESENT                 |
| Condition 3 | INTERVENTION | ABSENT                  |
| Condition 4 | INTERVENTION | PRESENT                 |

**Table 6:** The four conditions in an item set used in Experiment 2 and 3

## 4 Experiment 2 and 3: Does *wh*-informativity play a role in FIEs?

After clarifying the role of focus givenness in FIEs, we turn to the other potential pragmatic factor affecting FIEs pointed out in the literature—D-linking of *wh*-phrases. Given that simple D-linked *wh*-phrases have been tested in Jin and Yan (2024), the experiments reported in this section targeted complex D-linked *wh*-phrases, whose domains are further modified and thus more informative. We also used an acceptability-judgment task and a self-paced reading task to examine whether increased linguistic informativity ameliorates FIEs. A  $2 \times 2$  factorial design crossed the factors *structure* and *wh*-modification. To control for contextual effects, all stimuli were embedded in supporting contexts. The design details are presented in the following subsections.

### 4.1 Experiment 2: An acceptability judgment task

#### 4.1.1 Stimuli

The four conditions stemming from the  $2 \times 2$  design in Experiment 2 are illustrated in Table 6. *wh*-modification, a factor representing the informativity of a *wh*-phrase with the help of a relative clause modifier, has two levels—ABSENT, in which there is no modifier, and PRESENT, in which there is a modifier.

- (34) a. nǎ-gè lǎoshī  
 which-CLF teacher  
 ‘which teacher’ ABSENT
- b. [<sub>RC</sub> língdǎo biǎoyáng-gùo-de] nǎ-gè lǎoshī  
 leader praise-EXP-MOD which-CLF teacher  
 ‘which teacher praised by the leadership’ PRESENT

Adding a modifier increases the informativity of the *wh*-phrase by further specifying its domain. Moreover, it has been argued that a modifier contributes an additional specificity effect when attached to an indefinite phrase (Zhang 2006), as in [<sub>RC</sub> língdǎo biǎoyáng de] yí-gè lǎoshī ‘a teacher praised by the leadership’, which intuitively refers to a specific teacher known to the speaker (cf. von Heusinger 2002). D-linked *wh*-phrases with modifiers like (34-b) inquire about the identities of particular individuals.

*Structure* is a factor separating *wh*-questions that exhibit FIEs (INTERVENTION) from *wh*-questions without FIEs (PLAIN). Sample stimuli for the four experimental conditions are provided in (35)–(38). To facilitate the self-paced reading task, each PLAIN item was augmented with a pre-verbal modifier so that it matched the corresponding INTERVENTION item in length.

Condition 1: PLAIN + ABSENT (with a pre-verbal modifier)

- (35) Nà-gè xuéshēng [zài jiàoshījié xiàwǔ] sòng-gěi-le nǎ-gè lǎoshī  
that-CLF student on Teachers'.Day afternoon send-give-PFV which-CLF teacher  
yìbǎ xīnxiān-de báisè kāngnǎiqīng?  
one-CLF fresh-MOD white carnation  
'To which teacher did that student give a bouquet of fresh white carnations in the afternoon of Teachers' Day?'

Condition 2: PLAIN + PRESENT

- (36) Nà-gè xuéshēng sòng-gěi-le [<sub>RC</sub> lǐngdǎo biǎoyáng-gùo-de] nǎ-gè lǎoshī  
that-CLF student send-give-PFV supervisor praise-PFV-MOD which-CLF teacher  
yìbǎ xīnxiān-de báisè kāngnǎiqīng?  
one-CLF fresh-MOD white carnation  
'To which teacher praised by their supervisor did that student give a bouquet of fresh white carnations?'

Condition 3: INTERVENTION + ABSENT (with a pre-verbal modifier)

- (37) Zhǐyǒu NÀ-GÈ xuéshēng [zài jiàoshījié xiàwǔ] sòng-gěi-le nǎ-gè  
only that-CLF student on Teachers'.Day afternoon send-give-PFV which-CLF  
lǎoshī yìbǎ xīnxiān-de báisè kāngnǎiqīng?  
teacher one-CLF fresh-MOD white carnation  
'To which teacher did only that student give a bouquet of fresh white carnations in the afternoon of Teachers' Day?'

Condition 4: INTERVENTION + PRESENT

- (38) Zhǐyǒu NÀ-GÈ xuéshēng sòng-gěi-le [<sub>RC</sub> lǐngdǎo biǎoyáng-gùo-de] nǎ-gè  
only that-CLF student send-give-PFV supervisor praise-PFV-MOD which-CLF  
lǎoshī yìbǎ xīnxiān-de báisè kāngnǎiqīng?  
teacher one-CLF fresh-MOD white carnation  
'To which teacher praised by his supervisor did only that student give a bouquet of fresh white carnations?'

A reviewer noted that the stimuli used in this experiment are structurally complex and therefore may not sound entirely natural to native speakers. We share this concern; however, the complexity stems from the experimental design, which specifically targets linguistic informativity at the structural level—relative clauses embedded within the *wh*-phrases serve to enrich the informational content of their domains. Moreover, all items contained a complex expression following the *wh*-phrase, as they were also used in a subsequent self-paced reading task. This ensured comparability across tasks and provided a larger spill-over region which allows us to observe potential effects of FIEs on reading times. Although such complexity inevitably increases processing load, it was carefully controlled, as all minimal pairs were matched for structural complexity. In the discussion section, we will show that while this design led to a general degradation in acceptability, the effects of FIEs nonetheless remained detectable.<sup>11</sup>

<sup>11</sup> A reviewer pointed out that the postverbal region in the *wh*-modification-PRESENT condition may exhibit a confound garden path effect. In (36) and (38), specifically, the subject *lǐngdǎo* 'supervisor' of the relative clause is

By comparing these four conditions, we can measure the effect of *wh*-modification in *wh*-questions with and without FIEs to test the Linguistic Informativity Hypothesis, repeated in (39).

(39) **Linguistic Informativity Hypothesis**

- a. General version: Increased linguistic informativity of a *wh*-phrase facilitates the acceptability and processing of *wh*-in-situ questions generally, with or without FIEs.
- b. Specific version: Increased linguistic informativity of a *wh*-phrase has a specific ameliorating effect on FIEs.

If the general version is correct, we should expect higher acceptability ratings and faster reading times when *wh*-modification is present, regardless of FIEs. If the specific version is correct, we should expect higher acceptability ratings and faster reading times when FIEs and *wh*-modification are both present than when FIEs are present but *wh*-modification is absent.

All four conditions in an item set are paired with one and the same scenario, which ensures that any differences in the four conditions are not due to participants interpreting them relative to different contexts. More specifically, each scenario provides the following information: (i) the referent of the focused phrase, (ii) the alternatives to the focused phrase, (iii) the domain of the *wh*-phrase, and (iv) the modifier used in the test sentence. A sample scenario that goes with the sample stimuli is provided in (40).

(40) **Sample scenario** (English version)

Today is Teachers Day. A group of students were giving flowers to a group of teachers who have been praised by their supervisor. There was one particular teacher to whom only one student gave a bouquet of carnations. Xiaoming wanted to know who this teacher was, so he asked:

(41) A question is drawn from (35) - (38) as a stimulus.

most naturally parsed as the object of the main verb *sòng-gěi-le* ‘give to’, before the verb of the relative clause is parsed. However, this analysis is challenged by the additional verbal information *biǎoyáng-guò-de* ‘praised’, which indicates that *língdǎo* ‘supervisor’ should be reanalyzed as a subject rather than as an object.

Garden-path disambiguation typically increases first-pass reading times by roughly 50–150 ms, with total processing costs at the region often reaching several hundred milliseconds due to regressions and reanalysis (Frazier and Rayner 1982; Rayner 1998; Traxler et al. 1998; Zhou et al. 2021). There is no consensus on how much time it would take to recover from a garden path effect. According to Osterhout and Holcomb (1992), an ERP study, the recovery may take another 500ms to 900ms after the disambiguating word. However, studies also showed that recovery is not only costly, but can be incomplete (Ferreira and Henderson 1991).

We acknowledge that all of the stimuli with *wh*-modification-PRESENT are subject to a potential garden path effect. However, the reading times, as shown in (i), in the disambiguating region, RC2, and the region immediately following it, RC3, are not substantially different from regions preceding them, suggesting that any garden path effects, if present, are mild.

- (i) RC1 = *língdǎo* ‘supervisor’; RC2 = *biǎoyáng* ‘praise’; RC3 = *guò-de* PFV-MOD

| Region | structure    | wh-construction | RT (.ms) |
|--------|--------------|-----------------|----------|
| RC1    | INTERVENTION | PRESENT         | 200      |
| RC1    | PLAIN        | PRESENT         | 199      |
| RC2    | INTERVENTION | PRESENT         | 198      |
| RC2    | PLAIN        | PRESENT         | 191      |
| RC3    | INTERVENTION | PRESENT         | 197      |
| RC3    | PLAIN        | PRESENT         | 187      |

A total of 24 scenarios were created, yielding 96 target items across four conditions. The 96 items were further divided into four experiment lists using a Latin-Square design. A participant only read one of the four conditions in an item set. There were 24 target items in each list, along with 24 fillers. The 24 target items on each list would yield six items per condition per participant.

### 4.1.2 Procedure

The acceptability judgment task was designed and hosted on Ibex Farm (the version prior to PCibex) as a web-based study. The entire experiment was presented in Simplified Chinese. In the instruction page, a participant first read a short text describing the experimental procedure, before being asked to provide their language background, gender, age, and informed consent. This was followed by a practice phase, with three practice trials familiarizing a participant with the procedure of an acceptability judgment task. The experiment stage began after the practice session.

In the experiment phase, a participant first read a paragraph describing a situation, which provided the context for interpreting an upcoming *wh*-question. The text was presented all at once on the screen and did not time out. The participant then had to press the space bar to proceed to the test sentence after they were done reading the scenario. The test sentence was presented in full on the following screen.

The participant had to judge the acceptability of the sentence on a 7-point Likert scale by pressing the corresponding numbers on the keyboard. The lower end of the scale was 1, which was labeled in Simplified Chinese to mean ‘completely unacceptable’. The upper end of the scale was 7, which was labeled to mean ‘completely acceptable’. The mid-point, 4, was labeled to mean ‘marginal’.<sup>12</sup> A typical experimental session took about thirty minutes.

20 comprehension questions, 12 for target items and 8 for filler items, were used in this study to monitor each participant’s comprehension of the experimental material.

- (42) Xúesheng shì gěi lǎoshimen xiānhuā ma?  
 student be to teacher.PL give.flower Q  
 ‘Did the students give the teachers flowers?’ *Sample comprehension question*

All comprehension questions were in the form of polar questions. Half of them have a default ‘yes’ answer while another half a default ‘no’ answer. Two breaks, each lasting two minutes, were given in the experiment stage.

### 4.1.3 Participants

46 participants participated in Experiment 2. They were recruited from the Guangdong University of Foreign Studies ( $n = 40$ ) and Peking University Subject Pool ( $n = 6$ ). There were 36 female and 10 male participants. All identified themselves as native speakers of Mandarin. This experiment

<sup>12</sup> It should be noted that, unlike Experiment 1, Experiment 2 employed a 7-point scale instead of a 5-point scale. This choice was motivated by the greater structural complexity of the stimuli in Experiment 2, which warranted a finer-grained measure to capture potential nuanced interactions between structural complexity and FIEs. In addition, since cross-experiment comparisons were not intended, this difference in scale poses no methodological concern. That being said, we have accepted the suggestion, due to two reviewers, to use normalized (*z*-scored) ratings when reporting the results of Experiment 1 and Experiment 2.

|                      | $\beta$ | $SE$ | $t$    | $p$ |
|----------------------|---------|------|--------|-----|
| Intercept            | 0.00    | 0.08 | 0.00   |     |
| Intervention         | -.30    | .03  | -11.65 | *** |
| Present              | -.06    | .03  | -2.41  | *   |
| Intervention:Present | .06     | .03  | 2.17   | *   |

**Table 7:** Linear mixed-effects model of  $z$ -scored acceptability ratings with sum-coded **structure** (INTERVENTION) and **wh-modification** (PRESENT)

was conducted under the same IRB protocol and exempt determination described in the previous experiment.<sup>13</sup>

#### 4.1.4 Data treatment

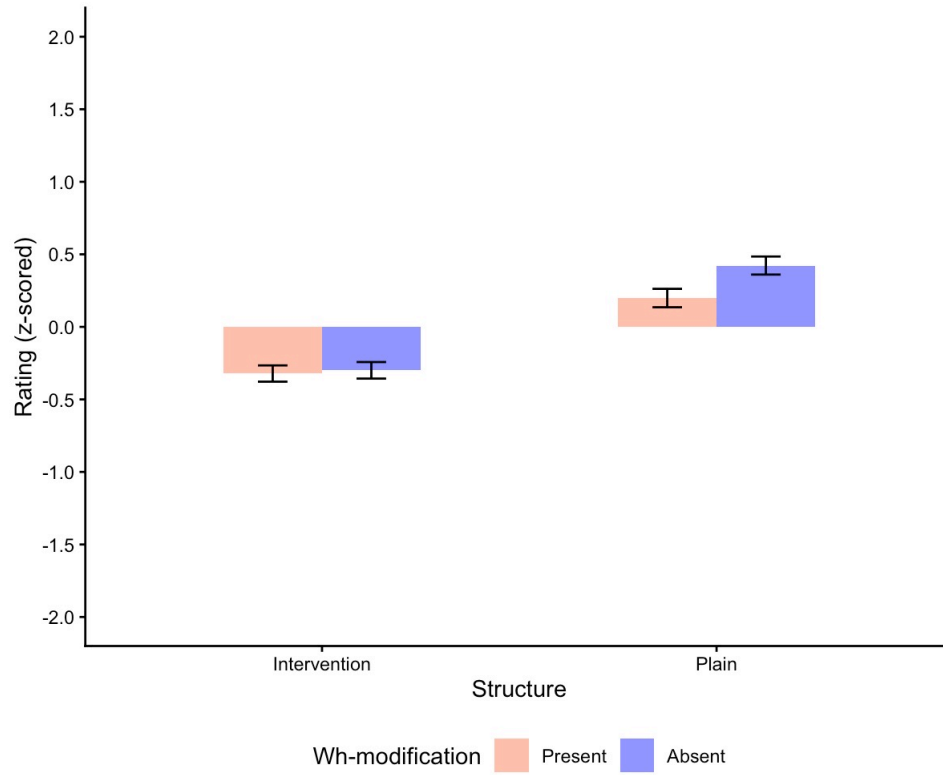
Prior to data analysis, 4 participants were excluded due to low comprehension accuracy rate—failing 8 or more comprehension questions out of 20. The total number of participants whose data remained in the data analysis was 42. They contributed 1,008 target data points, excluding fillers. A cumulative-linked mixed-effects model, from the R package *clmm*, was fit to examine the effects of **structure** (2 levels: PLAIN, INTERVENTION) and **wh-modification** (2 levels: ABSENT, PRESENT) on the dependent variable—**acceptability rating**, with participants and items as random intercepts. Both factors were coded using sum (deviation) coding, such that coefficients represent deviations from the grand mean rather than from a reference level.

#### 4.1.5 Results

Like Experiment 1, we transformed the ratings to  $z$ -scores. A linear model with sum-coded predictors for **structure** (INTERVENTION = +1, PLAIN = 1) and **wh-modification** (PRESENT = +1, ABSENT = 1) revealed a robust main effect of **structure**, such that responses were lower in the INTERVENTION than in the PLAIN condition ( $\beta = -0.30, SE = 0.03, t = -11.65, p < .001$ ). This contrast is also visible in Figure 6. There was also a significant main effect of **wh-modification**, with lower responses in the PRESENT than ABSENT condition ( $\beta = -0.06, SE = 0.03, t = -2.41, p = .02$ ). Importantly, these effects were qualified by a significant **structure**  $\times$  **wh-modification** interaction ( $\beta = 0.06, SE = 0.03, t = 2.17, p = .03$ ), indicating that the INTERVENTION-related decrease was attenuated in the PRESENT condition relative to the ABSENT condition. Similarly, these model results were confirmed by a cumulative-link (ordinal) mixed effect model based on raw ratings, as illustrated in Table 8.

Three conclusions can be drawn based on these results. First of all, **Experiment 2 argues against the general version of the Linguistic Informativity Hypothesis**. Increased informativity has a penalty effect on **wh**-in-situ dependencies, in contrast to the facilitation effect in filler-gap dependencies found in Hofmeister (2007). As pointed out by a reviewer, the baseline condition (i.e., PLAIN+ABSENT) received lower acceptability ratings than the corresponding baseline (i.e., PLAIN) in Experiment 1. This result is unsurprising, as acceptability judgments are partly affected by sentence length and syntactic complexity, independent of syntactic well-formedness (Wasow

<sup>13</sup> The recruitment procedures for Experiments 2 and 3 were identical to those used in Experiment 1.



**Figure 6:** Effects of **structure** and **wh-modification** on acceptability ratings (Experiment 1)

|                      | $\beta$ | $SE$ | $z$    | $p$ |
|----------------------|---------|------|--------|-----|
| Intervention         | -.69    | .06  | -11.19 | *** |
| Present              | -.16    | .06  | -2.83  | **  |
| Intervention:Present | .16     | .06  | 2.68   | **  |

**Table 8:** Cumulative-link mixed effects model of acceptability ratings with sum-coded **structure** (INTERVENTION) and **wh-modification** (PRESENT)

and Arnold 2005). The baseline condition in Experiment 2 was longer and structurally more complex due to a larger post-*wh* spill-over region. This is further supported by the comparable acceptability of the grammatical fillers, which match the baseline condition in structural complexity and length. The details are stored in Appendix E. Furthermore, the PLAIN+PRESENCE condition received even lower ratings for two reasons: it contained a relative clause, which is known to incur higher processing costs (Hsiao and Gibson 2003), and it further increased overall sentence complexity. Taken together, these factors account for why the PLAIN sentences without FIE patterns in Experiment 2 obtained a comparatively lower mean rating than its counterpart in Experiment 1.

Second, while both PLAIN conditions were somewhat degraded, **FIEs were again detected in this experiment**. The differences between the PLAIN conditions and the corresponding INTERVENTION conditions were still reliably detected by the statistical model. The significant main effect indicates that FIEs further contributed to the degradation of the INTERVENTION conditions. This pattern is consistent with the findings from Experiment 1 and with previous research.

Third, the interaction effect, which is reflected in the much smaller difference between the INTERVENTION conditions than between the PLAIN conditions, suggests **a potential advantage of increased informativity**, which is nonetheless attenuated by the additional complexity introduced by *wh*-modification. In other words, the processing cost associated with *wh*-modification may offset the informativity advantage. To further test this possibility, future research could manipulate informativity through alternative means, such as using shorter adjectives.

In short, **Experiment 2 does not produce insufficient evidence supporting the specific version of the Linguistic Informativity Hypothesis**, which states that FIEs can be ameliorated with the presence of increased informativity of a *wh*-phrase. Although there is an interaction effect between structure and *wh*-modification, this interaction merely results in the cancellation of the penalty induced by *wh*-modification in questions exhibiting FIEs, rather than an genuine amelioration effect.

## 4.2 Experiment 3: A self-paced reading task

### 4.2.1 Stimuli

The same stimuli used in Experiment 2 were also used in Experiment 3 but adapted for a self-paced reading paradigm. Each stimulus was segmented into 12 or 13 fragments, based on a combination of morphosyntactic and semantic considerations. For example, the two characters *nǎ* and *gè* belong in one fragment as *gè* is a bound morpheme. The two characters *xué* and *shēng* form a segment as they are parsed into a compound word. A segmented stimulus is provided in (43) and Table 9 as an example.

(43) Condition 4: Intervention + *wh*-modification present

S    **Zhǐyǒu** NÀ-GÈ xuéshēng zài jiàoshījié      xiàwǔ      sònggěi-le **nǎ-gè**  
      Only    that-CLF student    on Teachers' Day afternoon give-PFV    which-CLF  
**lǎoshī** yībǎ      xīnxiān-de báisè kāngnǎixīng?  
      teacher one-CLF fresh-MOD white carnation  
      'To which teacher did only that student give a bouquet of fresh white carnations in  
      the afternoon of Teachers' Day?'

Given the hypothesis that native speakers do not expect a *wh*-phrase to follow a focused phrase, the

| Region | Label | Segment            | Gloss         |
|--------|-------|--------------------|---------------|
| 1      | Foc-P | <i>zhǐyǒu</i>      | only          |
| 2      | Subj1 | <i>NÀ-GÈ</i>       | that          |
| 3      | Subj2 | <i>xúeshēng</i>    | student       |
| 4      | Adv1  | <i>zài</i>         | at            |
| 5      | Adv2  | <i>jiàoshījié</i>  | Teachers' Day |
| 6      | Adv3  | <i>xiàwǔ</i>       | afternoon     |
| 7      | Verb  | <i>sòng-gěi-le</i> | send-give-PFV |
| 8      | WH    | <i>nǎ-gè</i>       | which-CLF     |
| 9      | NP    | <i>lǎoshī</i>      | teacher       |
| 10     | NP+1  | <i>yìbǎ</i>        | one-CLF       |
| 11     | NP+2  | <i>xīnxiān-de</i>  | fresh         |
| 12     | NP+3  | <i>báisè</i>       | white         |
| 13     | NP+4  | <i>kāngnǎixīn</i>  | carnation     |

**Table 9:** A sample segmented stimulus in the INTERVENTION+ABSENT condition

critical region is Region 8, which contains the *wh*-item *nǎ-gè* which. If FIEs arise during online processing, we expect to observe longer reading times in this region when participants encounter the *wh*-phrase in (43). In addition to Region 8, we also examined the reading times of subsequent regions, since, as shown in Experiment 1 (see Section 3.5), a potential longer reading time may be detected in post-critical regions.

#### 4.2.2 Procedure

Experiment 3 featured a self-paced reading study, which was also designed and hosted on Ibex Farm. The procedure in Experiment 2 was the same as that in Experiment 2, with the exception of what is described as follows. Unlike Experiment 2, the experimental stimuli in Experiment 3 were presented fragment-by-fragment in a moving-window self-paced reading paradigm. Participants first saw a line of dashes on a computer screen and had to press the space bar to see the sentence in fragments. Their reading time for each fragment was recorded as they advanced through the sentence. The final region was followed by a question mark to mark the end of the sentence.

#### 4.2.3 Participants

Experiment 3 included 41 participants (10 males, 31 females), all recruited from the Guangdong University of Foreign Studies. All participants were native Mandarin speakers with competent reading abilities and were unaware of the study's purpose. This experiment was conducted under the same IRB protocol and exempt determination described in the previous experiments.

#### 4.2.4 Data treatment

Prior to data analysis, eight participants were excluded due to low comprehension accuracy (less than 66.7%). Also, any reading time that was slower than 100ms or longer than 1,000ms was ex-

| WH           | $\beta$ | $SE$ | $t$    | $p$ | NP           | $\beta$ | $SE$ | $t$    | $p$ |
|--------------|---------|------|--------|-----|--------------|---------|------|--------|-----|
| Intercept    | 2.47    | .02  | 112.75 | *** | Intercept    | 2.47    | .02  | 117.19 | *** |
| INTERVENTION | .00     | .01  | .43    |     | INTERVENTION | .00     | .01  | .54    |     |
| PRESENT      | -.00    | .01  | -.03   |     | PRESENT      | -.00    | .01  | -.64   |     |
| Interaction  | .03     | .01  | 1.72   | .   | Interaction  | -.00    | .01  | -.53   |     |

| NP+1         | $\beta$ | $SE$ | $t$       | $p$ | NP+2         | $\beta$ | $SE$ | $t$    | $p$ |
|--------------|---------|------|-----------|-----|--------------|---------|------|--------|-----|
| Intercept    | 2.47    | .02  | 124.53.20 | *** | Intercept    | 2.48    | .02  | 126.45 | *** |
| INTERVENTION | .00     | .00  | .74       |     | INTERVENTION | .00     | .00  | .69    |     |
| PRESENT      | .00     | .00  | .10       |     | PRESENT      | -.00    | .00  | -.57   |     |
| Interaction  | .00     | .00  | .08       |     | Interaction  | .00     | .00  | .60    |     |

**Table 10:** Linear mixed effects model showing of reading times with sum-coded `structure` and `wh-modification`

cluded.<sup>14</sup> This procedure affected about 20% of the total data points. There were 8,834 datapoints in the final data analysis. Self-paced reading times (log-transformed) were analyzed using linear mixed-effects models with sum-coded fixed effects of `structure` (INTERVENTION = +1, PLAIN = 1), `wh-modification` (PRESENT = +1, ABSENT = 1), and their interaction, fit separately for the WH, NP, NP+1, and NP+2 regions. Under this coding, the intercept represents the grand mean across conditions and contexts.

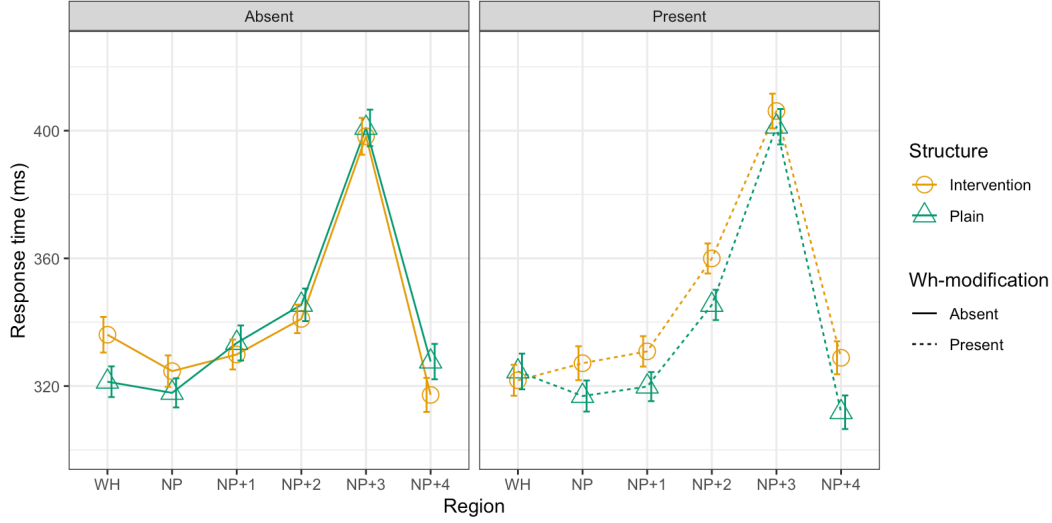
#### 4.2.5 Results

The results from Experiment 3 are reported in Figure 7. The left panel shows the reading times of plain and intervention sentences without `wh-modification` and the right panel the same sentences with `wh-modification`. Due to the length of the sentences involved, only the critical WH region and the spillover region (up to NP+4) is presented. The regions not shown are not directly comparable because the ABSENT condition and the PRESENT condition did not involve the same material: the ABSENT condition has a adverbial verbal modifier while the PRESENT condition has a nominal `wh-modifier`. The associated linear mixed-effects models (up to the NP+2 region) are reported in Table 10a–d.

Across all regions, there was no reliable main effect of `structure` (all  $|t| \leq 0.74$ ,  $ps > .45$ ) and no reliable main effect of `wh-modification` (all  $|t| \leq 0.64$ ,  $ps > .50$ ). At the WH region, the `structure`  $\times$  `wh-modification` interaction showed a marginal trend ( $\beta = 0.03$ ,  $SE = 0.01$ ,  $t = 1.72$ ,  $p = .085$ ), suggesting a weak tendency for the INTERVENTION effect to be attenuated in the PRESENT context; however, this interaction did not reach conventional significance. No interaction effects were observed at the NP, NP+1, or NP+2 regions (all  $|t| \leq 0.60$ ,  $ps > .55$ ). Overall, these results indicate that neither `structure` nor `wh-modification` exerted a robust influence on reading times in the critical region or in the spillover regions, with only a marginal interaction emerging at WH.

We can draw two conclusions from these results. First, a self-paced reading task is less sensitive

<sup>14</sup> A few different cutoff ranges were experimented with but they did not yield substantial differences. This particular range was chosen because of simplicity.



**Figure 7:** Effects of structure and *wh*-modification on reading times (Experiment 2; only the critical region and the spillover region are shown)

than an acceptability judgment task when deployed to detect FIEs in questions, confirming the observations from Experiment 1. It is important to understand why this asymmetry exists. Second, we have not observed any convincing evidence for the ameliorating effects of *wh*-modification, which is consistent with the findings in the acceptability judgment task. Note, however, that this conclusion must be qualified by the possibility that a self-paced reading task may not be fine-grained enough for detecting FIEs in the first place.

### 4.3 Interim summary

Based on two experiments with different dependent measures, we have learned a few things. Towards our goal of testing the Linguistic Informativity Hypothesis, we can conclude that there is no evidence, from the acceptability judgment task or the self-paced reading task, for any ameliorating effects of *wh*-modification. This conclusion stands in contrast to the observation that increased informativity facilitates the processing of filler-gap dependencies. We reserve this asymmetry for future research.

## 5 General discussion and theoretical outlook

The results from the three experiments together shed light on FIEs as well as on the typology of locality conditions more generally. As for FIEs, our results show that syntactic and semantic approaches to FIEs are still highly relevant. This is because the two prominent pragmatic factors we have examined, namely, focus givenness and linguistic informativity, do not eliminate FIEs. This does not mean that FIEs must be attributed to syntax or semantics. However, in the absence of other factors, syntax and semantic factors are the most well-understood ones so far in the literature.

That said, pragmatics itself is not a monolithic notion when it comes to modulating FIEs. Focus givenness does partially ameliorate FIEs, in contrast to *wh*-informativity, which shows no amelioration effect at all. This suggests that there is some empirical basis to the weaker version

of the Focus Givenness Hypothesis, adapted from Tomioka (2007) and Eilam (2011). The success of reducing some FIEs to focus givenness is encouraging. We think it is valuable to conduct more research to further decompose FIEs.

Moreover, the lack of sensitivity to *wh*-modification, as demonstrated in Experiment 2 and 3, offers unique evidence for comparing different types of interrogative dependencies. Given that Jin and Yan (2024)’s finding that informativity via D-linking also does not ameliorate FIEs, we can conclude that manipulating the linguistic informativity of an in-situ *wh*-phrase in Mandarin, whether through modification or D-linking, has no effect on FIEs. In this respect, FIEs in Mandarin, which involve *wh*-in-situ, contrast both with *wh*-movement and with *wh*-in-situ dependencies in English and French, which are sensitive to linguistic informativity (Pesetsky 1987, 2000; Hofmeister et al. 2013; Shlonsky et al. 2020, a.o.)

Taken together, sensitivity to linguistic informativity is only asymmetric across languages, but not within languages. In other words, in a *wh*-movement language, *wh*-phrases are sensitive to linguistic informativity, regardless of whether they are in-situ or moved. In a typical *wh*-in-situ language, *wh*-phrases are insensitive to linguistic informativity. To understand this cross-linguistic, rather than cross-construction difference, we suggest that Mandarin exhibits genuine *wh*-in-situ, while English (and possibly French) only involves pseudo-*wh*-in-situ.<sup>15</sup> If this suggestion is on the right track, then we can even go a step further and conclude that FIEs are not triggered by the failure of covertly moving an in-situ *wh*-phrase in syntax, as proposed in Beck (1996) and Beck and Kim (1997). This would then favor a syntactic account based on Agree (Kim 2006; Li and Cheung 2015), or a semantic account based on the interaction between focus and questions (Beck 2006; Mayr 2014; Li and Law 2016).<sup>16</sup>

Finally, the three experiments consistently show that FIEs are robustly detected in acceptability judgment tasks, but are substantially attenuated in self-paced reading when pragmatically facilitating factors are present. This pattern suggests that pragmatic facilitation may exert a stronger influence on online processing measures than on offline grammatical judgments. In the present experiments, pragmatic factors were examined in contexts where the relevant discourse information was explicitly provided. As a result, none of the lexical items in FIE sentences were discourse-new. Such contextual familiarity may facilitate lexical and incremental processing, thereby masking FIE-related processing difficulty in self-paced reading. Crucially, however, because FIEs reflect a grammatical constraint, they are not eliminated by contextual familiarity and re-emerge in offline acceptability judgments. Given that online processing tasks such as self-paced reading are comparatively difficult to design and implement, it remains an important open question how general this apparent insensitivity is, and addressing it will be crucial for guiding future experimental work on FIEs.

<sup>15</sup> Kotek (2014) argues that *wh*-in-situ in English can be derived through two mechanisms: covert movement and expansion of alternatives. Specifically, when an in-situ *wh*-phrase is stranded within an island, it takes scope through expansion of the alternatives it evokes, rather than through movement. As a consequence, an in-situ *wh*-phrase within an island exhibits FIEs. If true, we expect this kind of in-situ *wh*-expressions to also be insensitive to D-linking or informativity. This prediction remains to be tested in future research.

<sup>16</sup> A reviewer notes that, according to Bonan’s (2021) comparative study of Romance languages, French, particularly in contemporary varieties, may be undergoing a diachronic shift from *wh*-movement to *wh*-in-situ.

## 6 Conclusion

This paper presents three experiments designed to examine the role of pragmatics in FIEs, a configuration of which is given in (44). Although FIEs have been considered a kind of grammatical effect since their discovery in natural language, it has been hypothesized that some pragmatic factors can effectively mitigate or even eliminate such effects. This observation motivates the view that FIEs are, at least partially, pragmatic in nature. Our experiments targeted two specific pragmatic factors explicitly identified in the literature. These factors are formulated as two hypotheses, tested in the four experiments and summarized in (45) and (46).

- (44) FIE configuration  
[CP Q ... [IP only XP ... *wh*-phrase ] ... ]  
intervener
- (45) **Focus Givenness Hypothesis**
- Strong: If the focus phrase in a focus intervention sentence is contextually given, FIEs disappear.
  - Weak: If the focus phrase in a focus intervention sentence is contextually given, FIEs improve but may not entirely disappear.
- (46) **Linguistic Informativity Hypothesis**
- General version: Increased linguistic informativity of a *wh*-phrase facilitates the acceptability and processing of *wh*-in-situ questions generally, with or without FIEs.
  - Specific version: Increased linguistic informativity of a *wh*-phrase has a specific ameliorating effect on FIEs.

The experimental findings show asymmetric effects of pragmatics on FIEs. In particular, focus givenness improves FIEs but does not completely eliminate them; whereas linguistic informativity does not play an important role in FIEs. In light of these findings, we conclude that syntactic approaches and semantic approaches to FIEs are still viable and highly relevant. The lack of sensitivity to linguistic informativity further suggests that there may be two types of *wh*-in-situ: genuine *wh*-in-situ in predominately *wh*-in-situ languages like Mandarin and pseudo-*wh*-in-situ in predominantly *wh*-movement languages like English. We hope to further investigate this difference in future research.

## Supplementary files

Included in a separate document.

## Declarations

**Consent to Publish** Not applicable

**Consent to Participate** Informed consent was obtained from all individual participants included in the study.

**Ethics Approval** This study was approved by the the Institutional Review Board (IRB) at University of California, Santa Cruz (Approval No. HS3885). All procedures performed in studies

involving human participants were in accordance with the ethical standards of the institutional and/or national research committee.

**Data Availability** The data, experimental materials, and analysis scripts are publicly available at the Open Science Framework: <https://osf.io/eb4rm/overview>.

**Author Contribution** Jess H. K. Law: Writing – original draft, Supervision, Design and Conduct Experiment 2 & 3, Data analysis, Visualization, Conceptualization, Funding acquisition. Haoze Li: Writing – review, editing, revision, Resources, Design and Conduct Experiment 1, Data analysis, Conceptualization, Funding acquisition. Yaqing Cao: Design and conduct Experiment 2 & 3, Data analysis.

**Funding information** This work was supported by start-up grants provided by University of California, Santa Cruz and Nanyang Technological University.

## Acknowledgments

We are grateful to Ming Xiang, Lijuan Liang, Zheng Shen, Nick Huang, Matt Wagers, Jingnuo Chen, the audience of HSP-2022, the audience at the ELTS Seminar of National University of Singapore, and the audience at the TGIF Seminar of Nanyang Technological University for their comments and feedback. We are also indebted to the reviewers and the editor of Journal of East Asian Linguistics for their feedback and assistance.

## Competing interests

The authors have no competing interests to declare.

## References

- Aboh, E. O. (2007). Leftward focus versus rightward focus: The kwa-bantu conspiracy. In Kula, N. C. and Marten, L., editors, *Bantu in Bloomsbury: Special Issue on Bantu Linguistics*, pages 81–104. SOAS, University of London.
- Aoun, J. and Li, A. Y.-h. (1993). Wh-elements in situ: Syntax or LF? *Linguistic Inquiry*, 24(2):199–238.
- Beaver, D. and Clark, B. (2008). *Sense and Sensitivity: How Focus Determines Meaning*. Wiley-Blackwell.
- Beck, S. (1996). Quantified structures as barriers for LF movement. *Natural Language Semantics*, 4(1):1–56.
- Beck, S. (2006). Intervention effects follow from focus interpretation. *Natural Language Semantics*, 14(1):1–56.
- Beck, S. and Kim, S.-S. (1997). On wh- and operator scope in Korean. *Journal of East Asian Linguistics*, 6(4):339–384.

- Beck, S. and Kim, S.-S. (2006). Intervention effects in alternative questions. *Journal of Comparative Germanic Linguistics*, 9(3):165–208.
- Bonan, C. (2021). *Romance Interrogative Syntax: Formal and typological dimensions of variation*. John Benjamins Publishing Company, Amsterdam/Philadelphia.
- Cable, S. (2010). *The Grammar of Q: Q-Particles, Wh-Movement and Pied-Piping*. Oxford University Press.
- Chen, Z., Xu, Y., and Xie, Z. (2020). Assessing introspective linguistic judgments quantitatively. *Journal of East Asian Linguistics*, 29:311–336.
- Cheng, L. L.-S. (1991). *On the Typology of WH-Questions*. PhD thesis, Massachusetts Institute of Technology.
- Cheung, C. C.-H. (2014). Wh-fronting and the left periphery in Mandarin. *Journal of East Asian Linguistics*, 23:393–431.
- Chomsky, N. (1981). *Lectures on government and binding*. Foris Publications, Dordrecht, The Netherlands.
- Chomsky, N. (2005). Three factors in language design. *Linguistic Inquiry*, 36(1):1–22.
- Dong, H. (2009). *Issues in the semantics of Mandarin questions*. PhD thesis, Cornell University.
- Eilam, A. (2011). *Explorations in the Informational Component*. PhD thesis, University of Pennsylvania, Philadelphia, Penn.
- Erlewine, M. Y. (2025). Focus intervention, multiple association, and the unity of focus and *wh* alternatives. *Linguistics and Philosophy*.
- Erteschik-Shir, N. (1973). *On the Nature of Island Constraints*. PhD thesis, Massachusetts Institute of Technology.
- Falco, M. (2007). Weak crossover, specificity, and If chains. In Eguren, L. and Fern/’andez-Soriano, O., editors, *Coreference, modality, and focus*, pages 19–44.
- Ferreira, F. and Henderson, J. M. (1991). Recovery from misanalyses of garden-path sentences. *Journal of Memory and Language*, 30(6):725–745.
- Frazier, L. and Rayner, K. (1982). Making and correcting errors during sentence comprehension: Eye movements in the analysis of structurally ambiguous sentences. *Cognitive Psychology*, 14(2):178–210.
- Glasbergen, A. (2021). *Questions in context: the case of French wh-in-situ*. LOT dissertation series. LOT, Amsterdam.
- Goldberg, A. E. (2013). Backgrounded constituents cannot be extracted. In Sprouse, J. and Hornstein, N., editors, *Experimental syntax and island effects*, pages 221–238. Cambridge University Press.

- Hoeks, M., Toosarvandani, M., and Rysling, A. (2023). Processing of linguistic focus depends on contrastive alternatives. *Journal of Memory and Language*, 132:104444.
- Hofmeister, P. (2007). Memory retrieval effects on filler-gap processing. In *Proceedings of the Annual Meeting of the Cognitive Science Society*, pages 1091–1096.
- Hofmeister, P., Jaeger, F., Sag, I., Arnon, I., and Snider, N. (2013). The source ambiguity problem: Distinguishing effects of grammar and processing on acceptability judgments. *Language and Processes*, 28:48–87.
- Hoji, H. (1985). *Logical Form Constraints and Configurational Structures in Japanese*. PhD thesis, University of Washington, Seattle.
- Hsiao, F. and Gibson, E. (2003). Processing relative clauses in chinese. *Cognition*, 90:3–27.
- Huang, C.-T., Li, Y.-H. A., and Li, Y. (2009). *The Syntax of Chinese*. Cambridge University Press, Cambridge.
- Huang, C.-T. J. (1982). *Logical relations in Chinese and the theory of grammar*. PhD thesis, Massachusetts Institute of Technology.
- Ishihara, S. and Féry, C. (2006). Phonetic correlates of second occurrence focus. In *Proceedings of the 36th Annual Meeting of the North East Linguistic Society (NELS 36)*, Amherst, MA. GLSA.
- Jin, D. and Yan, H. (2024). Intervention effects in mandarin chinese—An experimental study. *Journal of Semantics*, pages 1–28.
- Just, M. A., Carpenter, P. A., and Woolley, J. D. (1982). Paradigms and processes in reading comprehension. *Journal of Experimental Psychology: General*, 111:228–238.
- Kim, S.-S. (2006). Question, focus, and intervention effects. In Kuno, S., editor, *Harvard Studies in Korean Linguistics XI*, pages 520–533.
- Kluender, R. and Kutas, M. (1993). Subjacency as a processing phenomenon. *Language and Cognitive Processes*, 8:573–633.
- Kotek, H. (2014). *Composing questions*. PhD thesis, Massachusetts Institute of Technology, Cambridge, MA.
- Kotek, H. and Erlewine, M. Y. (2016). Covert pied-piping in english multiple *wh*-questions. *Linguistic Inquiry*.
- Kuno, S. (1987). *Functional syntax: Anaphora, discourse and empathy*. Chicago University Press, Chocago/London.
- Li, A. Y.-H. (1992). Indefinite *wh* in mandarin chinese. *Journal of East Asian Linguistics*, 1(125155).
- Li, H. (2011). Focus intervention effects in mandarin. Master’s thesis, The Chinese University of Hong Kong, Hong Kong.

- Li, H. and Cheung, C. C.-H. (2015). Focus intervention effects in Mandarin multiple *wh*-questions. *Journal of East Asian Linguistics*, 24:361–382.
- Li, H. and Law, J. H.-K. (2016). Alternatives in different dimensions: A case study of focus intervention. *Linguistics and Philosophy*, 39:201–245.
- Liu, Y., Ryskin, R., Futrell, R., and Gibson, E. (2022). A verb-frame frequency account of constraints on long-distance dependencies in English. *Cognition*, 222.
- Lu, J., Thompson, C. K., and Yoshida, M. (2020). Chinese Wh-in-Situ and Islands: A Formal Judgment Study. *Linguistic Inquiry*, 51(3):611–623.
- May, R. (1977). *The Grammar of Quantification*. PhD thesis, Massachusetts Institute of Technology.
- Mayr, C. (2014). Intervention effects and additivity. *Journal of Semantics*, 31:513–554.
- Mitchell, D. C. (1984). An evaluation of subject-paced reading tasks and other methods of investigating immediate processes in reading. In Kieras, D. E. and Just, M. A., editors, *New Methods in Reading Comprehension Research*. Erlbaum, Hillsdale, N.J.
- Osterhout, L. and Holcomb, P. J. (1992). Event-related brain potentials elicited by syntactic anomaly. *Journal of Memory and Language*, 31(6):785–806.
- Pan, V. J. (2014). Wh-ex-situ in Chinese: Mapping between information structure and split CP. *Linguistic Analysis*, 39:371–413.
- Partee, B. H. (1999). Focus, quantification, and semantics-pragmatics issues. In Bosch, P. and van der Sandt, R., editors, *Focus: Linguistic, Cognitive, and Computational Perspectives*, pages 213–231. Cambridge University Press, Cambridge.
- Pesetsky, D. (1987). Wh-in-situ: Movement and unselective binding. In Reuland, E. J. and Meulen, A. G. B. t., editors, *The Representation of (In)definiteness*, pages 98–129. MIT Press, Cambridge, Massachusetts.
- Pesetsky, D. (2000). *Phrasal movement and its kin*. MIT Press, Cambridge, MA.
- Rayner, K. (1998). Eye movements in reading and information processing: 20 years of research. *Psychological Bulletin*, 124(3):372–422.
- Richards, N. (2001). *Movement in language: interactions and architectures*. Oxford University Press, Oxford.
- Rizzi, L. (1990). *Relativized minimality*. MIT Press, Cambridge, Massachusetts.
- Rizzi, L. (1997). The fine structure of the left periphery. In Haegeman, L., editor, *Elements of grammar*, pages 281–337. Kluwer Academic Publishers, Dordrecht.
- Rizzi, L. and Bocci, G. (2017). Left periphery of the clause: Primarily illustrated for Italian. In Everaert, M. and van Riemsdijk, H., editors, *The Blackwell Companion to Syntax*, pages 1–30. John Wiley and Sons, Hoboken, NJ.

- Roberts, C. (2012). Information structure in discourse: Towards an integrated formal theory of pragmatics. *Semantics & Pragmatics*, 5:1–69.
- Ross, J. (1967). *Constraints on variables in syntax*. PhD thesis, Massachusetts Institute of Technology.
- Shi, D. (1994). The nature of Chinese wh-questions. *Natural Language and Linguistic Theory*, 12(2):301–334.
- Shlonsky, U., Villata, S., and Franck, J. (2020). Covert movement in multiple-wh questions: Experimental and theoretical investigations. *Syntax*, 23(2):185–202.
- Soh, H.-L. (2005). Wh-in-situ in mandarin chinese. *Linguistic Inquiry*, 36(143-155).
- Sprouse, J., Caponigro, I., Greco, C., and Cecchetto, C. (2016). Experimental syntax and the variation of island effects in English and Italian. *Natural Language and Linguistic Theory*, 34:307–344.
- Sprouse, J., Schutze, C., and Almeida, D. (2013a). A comparison of informal and formal acceptability judgments using a random sample from Linguistic Inquiry 2001–2010. *Lingua*, 134:219–248.
- Sprouse, J., Wagers, M., and Phillips, C. (2012). A test of the relation between working-memory capacity and syntactic island effects. *Language*, 51(3):82–123.
- Sprouse, J., Wagers, M., and Phillips, C. (2013b). Deriving competing predictions from grammatical approaches and reductionist approaches to island effects. In Sprouse, J., editor, *Experimental syntax and island effects*, pages 21–41. Cambridge University Press.
- Tian, Q., Park, M.-K., and Yang, X. (2022). Mandarin chinese wh-in-situ argument-adjunct asymmetry in island sensitivity: Evidence from a formal judgment study. *Front Psychol*, 13:954175.
- Tomioka, S. (2007). Pragmatics of If intervention effects: Japanese and Korean WH-interrogatives. *Journal of Pragmatics*, 39:1570–1590.
- Tomioka, S., Kitagawa, Y., and Tamaoka, K. (2020). Intervention effects in Japanese revisited: Gradable acceptability and its theoretical consequences. In Barrie, M., editor, *Japanese/Korean Linguistics 27*, pages 19–36. CSLI Publications.
- Traxler, M. J., Pickering, M. J., and Clifton, Charles, J. (1998). Adjunct attachment is not a form of lexical ambiguity resolution. *Journal of Memory and Language*, 39(4):558–592.
- Tsai, D. W.-T. (1994). *On Economizing the Theory of A'-Dependencies*. PhD thesis, Massachusetts Institute of Technology.
- von Stechow, K. (2002). Specificity and definiteness in sentence and discourse structure. *Journal of Semantics*, pages 245–274.
- Wasow, T. (1972). *Anaphoric Relations in English*. PhD thesis, MIT, Cambridge, Massachusetts.

- Wasow, T. and Arnold, J. (2005). Intuitions in linguistic argumentation. *Lingua*, 115(11):1481–1496.
- Watanabe, A. (1991). Wh-in-situ, subjacency, and chain formation. In *MIT Occasional Paper in Linguistics* 2. MITPWL, Department of Linguistics and Philosophy, MIT.
- Wu, J. (1999). *Syntax and semantics of quantification in Chinese*. PhD thesis, University of Maryland, College Park.
- Xiang, M., Dillon, B., Wagers, M., Liu, F., and Guo, T. (2013). Processing covert dependencies: an SAT study on Mandarin *wh*-in-situ questions. *Journal of East Asian Linguistics*, 23:207–232.
- Xiang, M., Wang, S., and Cui, Y. (2015). Constructing covert dependencies: The case of Mandarin *wh*-in-situ dependency. *Journal of Memory and Language*, 84:139–166.
- Yang, B. C.-Y. (2008). *Intervention effects and the covert component of grammar*. PhD thesis, National Tsing Hua University, Taiwan.
- Yang, B. C.-Y. (2012). Intervention effects and *wh*-construals. *Journal of East Asian Linguistics*, 21(1):43–87.
- Zhang, N. (2006). Representing specificity by the internal order. *Linguistics*, 44:1–21.
- Zhou, P., Shi, J., and Zhan, L. (2021). Real-time comprehension of garden-path constructions by preschoolers: A mandarin perspective. *Applied Psycholinguistics*, 42(1):181–205.